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Environmental impact assessment for renewable energy investments through integrated reinforcement learning and molecular fuzzy-based decision-making algorithm

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ABSTRACT

Environmental impact assessment is a significant component of renewable energy project planning. However, the identification of key environmental performance indicators remains underexplored. In the literature, most existing studies do not adequately prioritize these environmental factors. This situation creates a significant research gap in the renewable energy literature. This study addresses this gap by proposing a novel hybrid decision-making model to identify the most effective investment strategies for improving the environmental performance of renewable energy projects. First, the balanced expert dataset has been constructed by Q-learning algorithm. The second stage is related to the evaluation of the criteria with molecular fuzzy (MF) Bayesian networks (BANEW)-based weighting. Finally, alternatives are ranked by MF multi-objective particle swarm optimization (MOPSO). This study fills an important gap in the literature on increasing the environmental sustainability of renewable energy investments by integrating molecular geometry-based fuzzy decision-making techniques and Q-learning supported expert weighting method in prioritizing environmental impacts. The use of molecular geometry and fuzzy multi-criteria decision-making analysis together reduces the uncertainty in the solution process of complex problems more effectively. The use of the Q-learning algorithm in the model reduces subjectivity in the decision-making process by providing a dynamic structure based on learning in the weighting of expert opinions. The findings show that biodiversity is the most effective environmental impact of renewable energy investments is mostly on biodiversity. On the other side, it is also identified that the most optimal option for assessing the environmental impact of renewable energy investments is life cycle assessment.

1. Introduction

Environmental impact assessment for renewable energy investments is important to minimize the potential impacts of these projects on the environment. Although renewable energy is clean, the way it is applied can harm nature. For example, wind turbines can negatively affect bird migration routes. Environmental impact assessment also increases the trust of local people and stakeholders in the project. Furthermore, this makes it possible for projects to be more compliant with legal regulations (Sharma et al, 2024). The protection of biodiversity is a fundamental criterion in environmental impact assessment. Especially large-scale projects such as hydroelectric power plants can affect natural habitats. Sustainable use of water resources is also a fundamental component of environmental impact assessment. Projects such as biomass and hydroelectricity have a direct impact on water consumption and water quality (Li et al., 2024). On the other hand, the impacts on air quality should also be evaluated during this process. This problem may occur during the installation stages of projects and transportation activities. Moreover, land use efficiency should also be analyzed within the scope of this assessment. Especially solar energy investments that require large areas can affect agricultural productivity.

Most studies in the literature address environmental impact criteria at a general level (Rabbani et al., 2024; Sekhar et al., 2024; Zhang et al., 2025). However, it has not been clearly established which criteria are

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more important in environmental impact assessments (Hall et al., 2024; Cui et al., 2025). In other words, the strategies that can be implemented in this process have not been prioritized with a systematic method (Gutsch and Leker, 2024; Sleiman and Ouf, 2025). These deficiencies lead to a series of serious problems in terms of ensuring environmental sustainability in renewable energy projects and making the right investment decisions (Gagliardi et al., 2025). Failure to determine these factors causes the environmental factors that are effective in projects to be ignored. This situation can lead to failure to make correct strategic decisions (Truong et al., 2025). This situation also causes decision makers to fail to develop a clear roadmap regarding which practices will be more effective. Furthermore, incorrectly determined environmental priorities lead to the waste of both economic and environmental resources. Considering these deficiencies, it is seen that a new priority analysis is needed. A new priority analysis will be an important step in making more sustainable and effective decisions.

Accordingly, the purpose of this study is to generate effective investment strategies to improve the environmental impact of the renewable energy projects. For this purpose, a new model is created that includes 3 stages and 26 steps. In the first stage, the balanced expert dataset has been constructed by Q-learning algorithm. The second stage is related to the evaluation of the criteria with molecular fuzzy (MF) Bayesian networks (BANEW)-based weighting. The final stage gives information about the ranking of the alternatives with MF multi-objective particle swarm optimization (MOPSO). The lack of consensus on which strategies are most effective is the main motivation for this study. This deficiency prevents investors from making the right decisions. Therefore, by conducting a new priority analysis, the right strategies can be presented to these decision makers. Thus, the main research questions of this study are demonstrated below.

(1) Which criteria for environmental impact assessment have the greatest weights?
(2) Which alternative strategies for assessing environmental impact in renewable energy investments should be prioritized?

This study contributes to the literature by determining prior investment strategies with a new model for the development of more environmentally sensitive renewable energy projects. This proposed model has some superiorities compared to the previously developed ones as detailed below. (1) The use of the Q-learning algorithm in the expert weighting process contributes significantly to the decision-making literature. In this context, expert weights are optimized according to changing conditions over time by learning from the knowledge of experts and using the reward/punishment mechanism. It prevents unbalanced opinions arising from differences between decision makers. Owing to this situation, the weights of the most effective experts can be determined objectively. (2) The use of molecular geometry and fuzzy multi-criteria decision-making analysis together increases the advantages of the proposed model. In this process, normalization is applied according to different molecular geometric shapes. Thus, the analysis results made with different shapes are used for comparison. This situation allows the uncertainty in the solution process of complex problems to be reduced. (3) Ranking of alternatives with MOPSO provides some advantages to the model. The correct solution is reached iteratively among these solutions with the Pareto principle. More realistic and accurate rankings are obtained due to the use of the optimization algorithm. Pareto optimality aims to find the best solutions that will not disrupt other criteria while improving one criterion.

This manuscript is organized as follows. Section 2 includes the evaluation of similar studies in the literature. With the help of this evaluation, a criteria list is created. Additionally, this evaluation provides an opportunity to identify the missing part in the literature. Section 3 gives information about the steps of the proposed model that is created to satisfy this missing area of renewable energy literature. Section 4 demonstrates the main findings of this proposed model. Within this framework, criteria weights and alternative rankings can be identified. Section 5 denotes the comparison of these findings with the results of previous similar studies. Section 6 indicates the main conclusion, limitations and future research recommendations.

2. Literature review

Renewable energy investments are needed for more sustainable management of biodiversity (Kwilinski, 2024). Investments in renewable energy help protect biodiversity by reducing carbon emissions (Aydin et al., 2024). Compared to traditional energy production methods, renewable energy is important for lower environmental destruction (Arshed et al., 2024). In addition, the extraction of fossil fuels causes more damage to natural habitats than renewable energy (Acker et al., 2023). Qing et al. (2024) examined the relationship between renewable energy investments carbon neutrality and economic sustainability. The results of the study carried out in the Asian region argue that the use of green energy protects biodiversity. Yan et al. (2023) conducted a study on taking sustainable steps to promote the use of clean energy. It is suggested that using renewable energy can contribute to the conservation of biodiversity as it can reduce the emission of greenhouse gases.

Renewable energy investments contribute to more efficient use of water resources (Noh et al., 2024). Energy generated by nuclear power plants and fossil fuels can use huge amounts of water for cooling (Dagar et al., 2024). In addition, renewable energy sources such as solar and wind do not require water use in the production process (Bukhary, 2024). Therefore, more efficient use of water is associated with the use of renewable energy. Jafarinejad et al. (2023) examined the relationship between energy, water, and the environment in California. It is stated that the use of renewable energy provides significant benefits in water consumption as well as benefits such as greenhouse gas reduction and positive contribution to climate change. Adedoyin (2024) critically examined policies on the relationship between renewable energy, water, and the environment. Interviews with key stakeholders show that energy and water policies need to be built greener.

Renewable energy use has a positive impact on air quality improvement (Xie et al., 2023). Reducing carbon emissions is directly related to the use of renewable energy (Kartal et al., 2024). Fossil fuels are one of the main causes of carbon dioxide and global warming (Chen et al., 2023). Furthermore, burning fossil fuels to produce energy can cause acid rain and respiratory diseases (Sivalingam et al., 2024). However, the use of renewable energy minimizes these issues and allows for improved air quality (Soto, 2024). Qin et al. (2024) conducted a study to evaluate China's carbon neutrality target. It is argued that the use of renewable energy is an important tool to achieve carbon neutrality. Akça (2024) examined the impact of renewable energy sources on air quality in developed and developing countries. It is stated that the use of renewable energy has a positive effect on air quality even in developed industries.

Land productivity is an issue related to investments in renewable energy (Ahakwa, 2024). The extraction and processing of fossil fuels cause harmful pollutants to migrate into the land. This is one of the main reasons for declining land productivity (Soni & RL, 2024). In addition, renewable energy sources such as hydropower have a positive impact on the regular irrigation and productivity of agricultural land (Naimoglu et al., 2024). Moreover, the minerals produced in the geothermal energy production process increase land productivity (Wang and Zhang, 2023). Therefore, it can be said that renewable energy investments have a positive contribution to land productivity. Basso and Zolin (2023) examined the land productivity of farms producing renewable energy. The results of the study conducted in Italy show that the performance of farms that produce renewable energy is better than those that do not. Edouard et al. (2023) aim to improve the land productivity of alfalfa fields with photovoltaics. It is seen that the use of photovoltaics contributed to an increase of 51 % in the production process.

As a result of the literature review, some important points stand out. Primarily, the popularity of renewable energy investments is increasing due to their impact on reducing carbon emissions and combating climate change. However, the environmental impacts of renewable energy investments are still a matter of debate. Biodiversity, efficient use of water resources, minimizing air pollution, and improving soil quality are all associated with renewable energy investments. This shows the need for a new study that identifies the most important indicators. However, there are few studies in the literature that consider this issue. In this study, a priority analysis was conducted to fill this gap in the literature.

3. Methodology

In this paper, where environmental impacts for renewable energy investments are evaluated, an integrated reinforcement learning, and MF decision-making algorithm is proposed for the analysis process. In this analysis process, the criteria determined for environmental impact assessment are weighted with the MF BANEW method. In the next step of the process, alternative strategies are developed to assess the environmental impact in renewable energy investments, and the optimal strategy is selected with the help of the MF MOPSO method. In these stages, molecular fuzzy sets (MFSs) introduced in section 3.1 are used. In addition, Q-learning algorithm for unbalanced decision maker evaluations is used to obtain balanced evaluations of decision makers. The visual presentation of the analysis process is depicted in Fig. 1.

3.1. MFS

Inspired by molecular geometry shapes, MFS can produce more balanced, higher accuracy and more proportional results in solving complex problems under uncertainty. In addition, solutions calculated for different degrees of molecular geometry shapes can be used for comparison (Gentili, 2024). This allows the consistency of the results to be tested. For MFSs (A) in a universe of discourse (S), $a_i = (\beta_i, \gamma_i, \delta_i)$ is defined on condition that $\beta_A(a) + \gamma_A(a) + \delta_A(a) = 1$. Here β is the membership degree and is formulated by Eq. (1).

$$\beta_A(a) = f(\omega_a) = 1 - \frac{\omega_a}{\omega_{max}} \tag{1}$$

Where ω_a means the normalized angle of the membership degrees for item a, $f(\omega_a)$ is the monotonically decreasing function of the normalized angles in a range [0,1]. ω_{max} shows the maximum normalized angle value or the angles of the selected molecular geometry shapes in the matrix. According to the selected molecular geometry shapes, ω_{max} can be 180, 120, 109.5, 105, 90 degrees, or the radian equivalents of these angles. According to the order of the angles, it is named as linear shape, trigonal planar shape, tetrahedral shape, trigonal bipyramidal shape and octahedral shape. The other component in the condition of MFS, the degree of non-membership (γ), is defined as in Eq. (2).

$$\gamma_A(a) = \frac{\omega_a}{\omega_{max}} \tag{2}$$

The hesitant degrees of the MFS are calculated with the help of Eq. (3).

$$\delta_A(a) = 1 - \left(\left(1 - \frac{\omega_a}{\omega_{max}}\right) + \frac{\omega_a}{\omega_{max}}\right) \tag{3}$$

Since the Eq. (3) includes the angle and the normalization of the angle according to molecular geometric shapes, the angular converges to zero. In addition, the concepts of membership and non-membership are fuzzy numbers and involve different calculations than real numbers.

3.2. Q-learning algorithm for unbalanced decision Maker's evaluations

The depth of the collected evaluations varies according to the quality of the decision makers. In other words, the more experienced and knowledgeable the decision maker is, the more important the decision maker's evaluations become. Therefore, in multi-criteria decision-making methods, the evaluations of other decision makers should be balanced according to the evaluations of the best decision maker (Zhou et al., 2024). Due to professional factors such as experience, the knowledge of decision makers is not of the same importance. Therefore, the evaluations of other decision makers should be balanced according to the most experienced decision maker. The Q-learning algorithm is an iterative method that updates the matrices of other decision makers based on the evaluation of the most experienced decision makers.

In this context, the Q-learning algorithm is one of the most unique methods. Within the scope of the Q-learning algorithm, the reward $(\Theta_{s,a})$ and penalty $(\Gamma_{s,a})$ degrees between the best decision maker and the others are calculated with the help of Eqs. (4) and (5), and the K values are updated using Eq. (6) for different learning rates.

$$\Theta_{s,a} = \theta \cdot \left(\mathrm{K}_{s,a(best)} - \mathrm{K}_{s,a(others)}\right) \tag{4}$$

$$\Gamma_{s,a} = \zeta \cdot \left(\mathrm{K}_{s,a(others)} - \mathrm{K}_{s,a(best)}\right) \tag{5}$$

$$\mathrm{K}'_{s,a} = \mathrm{K}_{s,a(best)} + \iota \cdot \left(\Theta_{s,a} - \Gamma_{s,a}\right) \tag{6}$$

Wherein $\mathrm{K}_{s,a(best)}$ is the evaluation of the best decision maker and $\mathrm{K}_{s,a(others)}$ is the evaluation of the others. θ, ζ and ι are the reward, penalty factors and learning rate. The optimal K value is obtained when the maximum absolute difference (δ_{max}) between the initial and update values is less than the threshold value. For this, the maximum of difference in Eq. (7) is calculated with Eq. (8).

$$\delta_{s,a} = \left|\mathrm{K}'_{s,a} - \mathrm{K}_{s,a}\right| \tag{7}$$

$$\delta_{max} = \max_{s,a}\left\{\delta_{s,a}\right\} \tag{8}$$

Finding the optimal value is an iterative process and continues until the result of Eq. (8) is lower than the threshold value. Thus, the evaluations of the most experienced decision maker and other decision makers are ensured to be similar. These matrices obtained are ensured to be of the quality of the best decision maker, as well as the decision makers' evaluations. When the termination condition of the iteration is met, the obtained K values are the final K values.

Each state $s \in S$ corresponds to a specific pairwise comparison between the best or the most weighted decision maker DM_1 and another decision maker DM_k. Where $s = (\mathrm{DM}_1, \mathrm{DM}_k)$, k is k^{th} decision maker, S defines the state space. Action space (A) for each state s, an action $a \in A$ donates applying the Q-learning update rule to adjust the fuzzy relation and decision matrices entry for that DM pair. Reward function (Γ) is defined upon taking action a in state s. The immediate reward $\Gamma_{s,a}$ is computed as the reward degree θ equal to the weight of the less experienced decision maker in the pairwise comparison. Penalty factor ζ is the weight of the most experienced decision maker. State transition (P) is defined after updating $\Theta_{s,a}$, the algorithm moves to the next state. Once all pairs are updated, one learning iteration completes and the next begins, repeating until convergence.

3.3. MF-BANEW

The MF-BANEW method is a multi-criteria decision-making technique that weights the criteria. Thus, the conditional dependencies or interactions between the criteria are considered during the calculation of weights. In this way, more accurate results can be obtained thanks to Entropy, Information Gain and Conditional Probability calculations. The basic steps of the method are as follows (Wang et al., 2024).

Degrees of dependency between criteria are assigned by decision makers. The degree of dependence of the criteria is transformed into MFSs. The average of the MFSs of each decision maker's evaluation is calculated by Eq. (9).

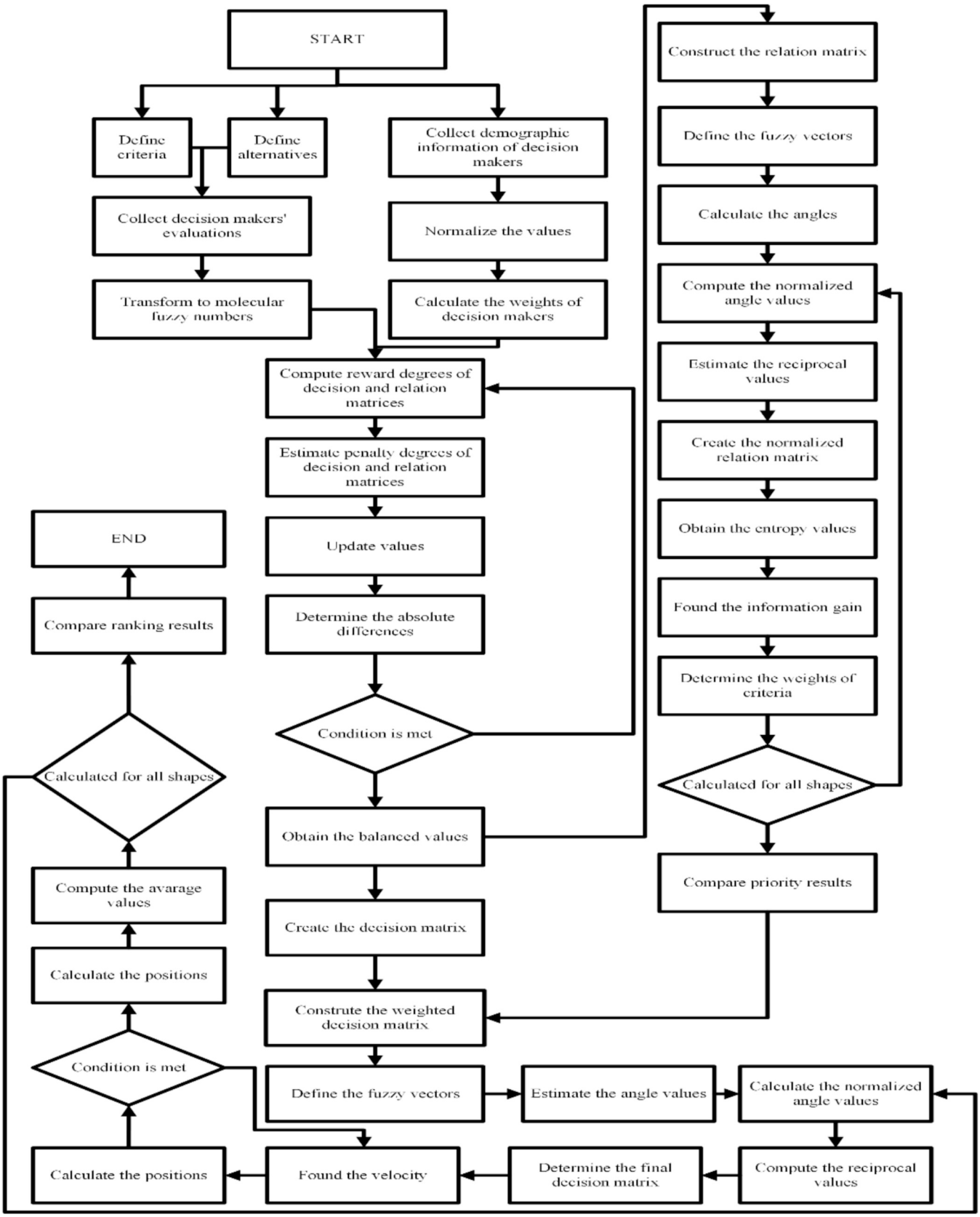

Fig. 1. The Visual presentation of the analysis process.

$$\mathscr{C} = \left(\bigcup_{i=1}^{e} \mathscr{C}_i\right) = \left\{ \left(a, \frac{1}{e}\sum_{i=1}^{e} \beta_{\mathscr{C}_i}(a), \frac{1}{e}\sum_{i=1}^{e} \gamma_{\mathscr{C}_i}(a), \frac{1}{e}\sum_{i=1}^{e} \delta_{\mathscr{C}_i}(a) \right) \middle| a \in A \right\} \tag{9}$$

The relation matrix ($\mathscr{C}$) is constructed with the average of the evaluations of decision makers. The number of decision makers is e. Afterwards, fuzzy vectors are defined with other relations except the relation between the criteria themselves. The fuzzy vectors ($\mathscr{L}$) are created with the help of Eq. (10).

$$\mathscr{L}_i = \left[(\beta_{i1}, \gamma_{i1}, \delta_{i1}), (\beta_{i2}, \gamma_{i2}, \delta_{i2}), \cdots \left(\beta_{i(n-1)}, \gamma_{i(n-1)}, \delta_{i(n-1)}\right) \right] \tag{10}$$

Using fuzzy vectors, the angle values ($\mathscr{o}$) shown in Eq. (11) are calculated. In this process, dot product, cosine similarity, inverse cosine function and magnitude of fuzzy vector calculations are used.

$$\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j} = cos^{-1}\left(\frac{\sum_{t=1}^{n-1}\left(\beta_{i,t}\cdot\beta_{j,t} + \gamma_{i,t}\cdot\gamma_{j,t} + \delta_{i,t}\cdot\delta_{j,t}\right)}{\left(\sum_{t=1}^{n-1}\left(\beta_{i,t}^2 + \gamma_{i,t}^2 + \delta_{i,t}^2\right)\right)\cdot\left(\sum_{t=1}^{n-1}\left(\beta_{j,t}^2 + \gamma_{j,t}^2 + \delta_{j,t}^2\right)\right)} \right) \tag{11}$$

In the normalization phase of the angles, the general situation is to divide the angles by maximum value. This situation is shown in Eq. (12).

$$\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)_{gen} = \frac{\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}}{\mathscr{o}_{max}} \tag{12}$$

Wherein $\aleph$ is normalized angles. However, due to the advantage of the MFSs, normalization is performed by taking advantage of different geometry shapes. Eq. (13) defines the normalizations corresponding to different molecular geometry shapes.

$$\begin{aligned}
\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)_{linear} &= \frac{\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}}{\pi} \\
\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)_{tripl} &= \frac{\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}}{\frac{2\pi}{3}} \\
\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)_{tet} &= \frac{\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}}{\frac{\pi}{2}} \\
\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)_{tribp} &= \frac{\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}}{\frac{2\pi}{5}} \\
\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)_{oct} &= \frac{\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}}{\frac{\pi}{3}}
\end{aligned} \tag{13}$$

Then, the reciprocal values ($\Im$) of the normalized angles are computed using Eq. (14).

$$\Im\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right) = \frac{1}{\aleph\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)} \tag{14}$$

With Eq. (15), the reciprocal values are normalized, and the normalized matrix of the relation matrix is constructed with these normalized reciprocal values. The normalization of the relation matrix ($\mathscr{H}$) is shown by Eq. (16).

$$\hbar_{ij} = \frac{\Im\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)}{\sum_{j=1}^{n}\Im\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)} \tag{15}$$

$$\mathscr{H} = \begin{bmatrix} 0 & \hbar_{12} & \cdots & \cdots & \hbar_{1n} \\ \hbar_{21} & 0 & \cdots & \cdots & \hbar_{2n} \\ \vdots & \vdots & \ddots & \cdots & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \hbar_{n1} & \hbar_{n2} & \cdots & \cdots & 0 \end{bmatrix} \tag{16}$$

After normalization of the relation matrix, the entropy ($E(C_i)$) and conditional probability entropy ($E(C_i|C_j)$) values of the criteria are obtained with the help of Eqs. (17) and (18), respectively.

$$E(C_i) = -\sum \wp(C_i).In(\wp(C_i)) \tag{17}$$

$$E(C_i|C_j) = -\sum \wp(C_j).\wp(C_i|C_j).In(\wp(C_i|C_j)) \tag{18}$$

The difference between two entropy values is defined as information gain ($Gain$) and is calculated by Eq. (19).

$$Gain(C_i) = E(C_i) - E(C_i|C_j) \tag{19}$$

Finally, the weights ($\mathscr{w}$) of the criteria are estimated by Eq. (20).

$$\mathscr{w}(C_i) = \frac{Gain(C_i)}{\sum_{e=1}^{n} Gain(C_e)} \tag{20}$$

3.4. MF-MOPSO

MF-MOPSO produces balanced results by establishing pareto-optimal solutions to complex multi-objective decision-making problems. More accurate and comprehensive rankings are obtained thanks to the interaction of alternatives in the same order and spatial relations. The MF-MOPSO's computing steps are given in below (Gentili, 2024).

Linguistic evaluations are collected, transformed into MFSs, and the MF decision matrix is constructed using Eq. (9). Afterwards, the values of the MF decision matrix are multiplied by the criteria weights as shown in Eq. (21).

$$WD_{ij} = \left(\mathscr{w}_j.\beta_{ij}, \mathscr{w}_j.\gamma_{ij}, \mathscr{w}_j.\delta_{ij}\right) \tag{21}$$

Wherein WD_{ij} is the element of weighted decision matrix. Then, fuzzy vectors (Ω) are defined for each alternative. Fuzzy vectors are determined in Eq. (22).

$$\Omega_i = [(\beta_{i1}, \gamma_{i1}, \delta_{i1}), (\beta_{i2}, \gamma_{i2}, \delta_{i2}), \cdots (\beta_{im}, \gamma_{im}, \delta_{im})] \tag{22}$$

With fuzzy vectors, the angle values ($\mathscr{g}$) shown in Eq. (23) are computed. In this process, dot product, cosine similarity, inverse cosine function and magnitude of fuzzy vector calculations are applied.

$$\mathscr{g}_{\Omega_i,\Omega_j} = cos^{-1}\left(\frac{\sum_{t=1}^{n}\left(\beta_{i,t}\cdot\beta_{j,t} + \gamma_{i,t}\cdot\gamma_{j,t} + \delta_{i,t}\cdot\delta_{j,t}\right)}{\left(\sum_{t=1}^{m}\left(\beta_{i,t}^2 + \gamma_{i,t}^2 + \delta_{i,t}^2\right)\right)\cdot\left(\sum_{t=1}^{m}\left(\beta_{j,t}^2 + \gamma_{j,t}^2 + \delta_{j,t}^2\right)\right)} \right) \tag{23}$$

In the normalization phase of the angles, the general situation is to divide the angles by maximum value. This situation is shown in Eq. (24).

$$\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)_{gen} = \frac{\mathscr{g}_{\Omega_i,\Omega_j}}{\mathscr{g}_{max}} \tag{24}$$

However, due to the advantage of the MFSs, normalization is performed by taking advantage of different geometry shapes. Eq. (25) defines the normalizations corresponding to different molecular geometry shapes.

$$\begin{aligned}
\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)_{linear} &= \frac{\mathscr{g}_{\Omega_i,\Omega_j}}{\pi} \\
\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)_{tripl} &= \frac{\mathscr{g}_{\Omega_i,\Omega_j}}{\frac{2\pi}{3}} \\
\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)_{tet} &= \frac{\mathscr{g}_{\Omega_i,\Omega_j}}{\frac{\pi}{2}} \\
\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)_{tribp} &= \frac{\mathscr{g}_{\Omega_i,\Omega_j}}{\frac{2\pi}{5}} \\
\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)_{oct} &= \frac{\mathscr{g}_{\Omega_i,\Omega_j}}{\frac{\pi}{3}}
\end{aligned} \tag{25}$$

Then, the reciprocal values of the normalized angles are calculated with the help of Eq. (26).

$$\Im\left(\mathscr{g}_{\Omega_i,\Omega_j}\right) = \frac{1}{\aleph\left(\mathscr{g}_{\Omega_i,\Omega_j}\right)} \tag{26}$$

Using Eq. (27), the reciprocal values ($\Im$) are normalized, and the final decision matrix (η) is constructed with these normalized reciprocal values.

$$\eta_{ij} = \frac{\Im\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)}{\sum_{j=1}^{n}\Im\left(\mathscr{o}_{\mathscr{L}_i,\mathscr{L}_j}\right)} \tag{27}$$

Afterwards, MOPSO algorithm process is performed. In this context, the particle presentation for the potential solutions of the final decision matrix is obtained by Eq. (28).

$$\text{Ç}_i = \{\varsigma_{i1}, \varsigma_{i2}, \cdots, \varsigma_{in}\} \tag{28}$$

Then, the velocity (V) of each particle is estimated with the help of Eq. (29).

$$V_{ij}(t+1) = 0.5V_{ij}(t) + 1.5x_1\left(S_{ij}(t) - \text{Ç}_{ij}(t)\right) + 1.5x_2\left(S_{gb_j}(t) - \text{Ç}_{ij}(t)\right) \tag{29}$$

Where, x_1 and x_2 are random numbers between 0 and 1. Also, t represents the iteration step. After that, Eq. (30) gives information about the initial velocity.

$$V_{ij}(1) = 0.1\left(S_{max_i} - S_{min_i}\right)r \tag{30}$$

Where, r is random value between −1 and 1. After the initial velocity is calculated, the position of each particle is updated with Eq. (31).

$$\text{Ç}_{ij}(t+1) = \text{Ç}_{ij}(t) + V_{ij}(t+1) \tag{31}$$

The process of updating the values continues until the condition in Eqs. (32)–(34) is met.

$$S_{ij}(t+1) = Best\left(S_{ij}(t), \text{Ç}_{ij}(t+1)\right) \tag{32}$$

$$S_{gb_j}(t+1) = Best\left(S_{gb_j}(t), \{\text{Ç}_{ij}(t+1)\}alli\right) \tag{33}$$

$$\left|S_{gb_j}(t+1) - S_{gb_j}(t)\right| < 0.001 \tag{34}$$

When the condition is met and the iteration is completed, the personal and global best positions in final decision matrix are obtained. Finally, average values are obtained using Eq. (35) and the ranking of the alternatives (Λ) is carried out according to this value.

$$\Lambda_j = \frac{1}{n}\sum_{i=1}^{n} S_{ij} \tag{35}$$

4. Analysis results

4.1. Problem description

Renewable energy investments are an important alternative to reduce the environmental impacts of fossil fuels. This is also necessary for countries to achieve their sustainable development goals. The important issue in this process is that these investments should be planned in a way that is sensitive to nature. Otherwise, these projects may cause environmental damage. Many studies in the literature address environmental impact assessment criteria in general terms. However, it is a significant deficiency in the literature that it is not clear which criteria are more critical. This increases the risk of not taking environmental priorities into consideration sufficiently when making strategic decisions. This situation causes both economic and environmental resources to be used ineffectively. In summary, in this process, the most critical variables need to be determined so that businesses can use their limited resources effectively.

The different numbers of learning rate (ι) are selected as $\iota \in \{0.1, 0.5, 1\}$ respectively and tested for measuring the coherency of the proposed Q-learning methodology. The results show that the different learning rates have similar ranking performances with the best balance between stability and speed of convergence in Q-value updates. The reward factor θ for each DM pair is set to weight of the less experienced DM (0.300 for DM_2 and 0.285 for DM_3), the penalty factor ζ equals to the weight of the best DM (0.415 for DM_1) as defined in –Table A4. Convergence threshold ε is set as 0.02 and it is constructed by the iterations until $\delta_{max} < \varepsilon$. If the maximum difference among the Q values is less than the convergence threshold, the stability is provided for the final Q values based on the trends in Tables A11–A17. However, a maximum of 50 iterations is allowed in the definition of the convergence in the relation and decision matrices. Regarding the MF-MOPSO parameters, inertia weight defining the exploration capacity, cognitive, and social coefficients, and maximum velocity are assigned as 0.5, 1.5, 1.5, 0.001 respectively. The random values are limited between 0 and 1. However, the training process of the Q-learning algorithm is initiated by using Q-matrix evaluations from the most experienced decision maker (DM_1), the episode loops are determined by using the trained iterations are reached after 5 iterations for the relation matrix and 4 iterations for the decision matrix.

4.2. Creating the balanced decision maker dataset with Q-learning algorithm

MFSs are used in the analyses conducted for environmental impact assessment for renewable energy investments. In this context, the linguistic scales equivalents constructed based on Equations (1)-(3) are visualized in Fig. 2.

The criteria chosen for the environmental impact assessment are biodiversity (BIODIV), water resources (WTRRES), air quality (AIRQUA) and land productivity (LNDPRD). Biodiversity ensures that ecosystems function healthily. Loss of biodiversity leads to the deterioration of general environmental services. This situation negatively affects the sustainability of the project in the long term (Dias et al., 2025). Water resources are one of the indispensable criteria for environmental impact assessment. Water pollution can occur especially during the construction phase of these projects (Ji et al., 2025). Air quality plays a fundamental role in environmental impact assessments. Fuel consumption during the construction process of projects can negatively affect air quality (Wang et al., 2025). Land productivity is an important criterion within the scope of environmental impact assessment. Infrastructure works carried out in large-scale projects can disrupt the soil structure (Kyttä et al., 2025).

Similarity, the alternative strategies selected for assessing environmental impact in renewable energy investments are life cycle assessment (LFCYAS), dynamic systems over time (DYSYSOT), geographic information systems (GEOINFSYS), ecosystem services (ECOSRV) and environmental performance (ENVPER). Life cycle assessment examines

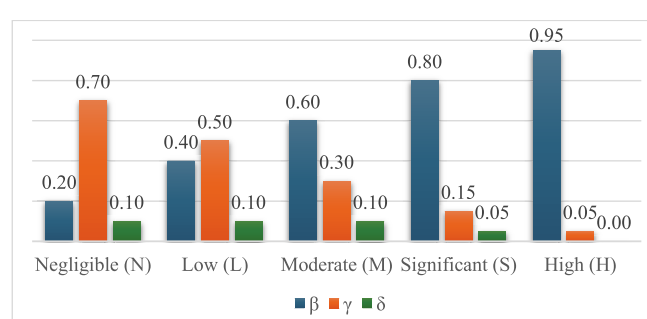

fig. 2. linguistic scales.

environmental impacts such as energy use at each stage (Su et al., 2025). With the help of this condition, it is possible to objectively compare the environmental performance of different energy technologies. Dynamic systems provide an understanding of how the interaction of different factors changes over time (Shanbhag and Dixit, 2025). Geographic information systems are used to understand where and how environmental, social and economic impacts occur geographically (El-Sorogy et al., 2025). On the other hand, ecosystem services analyze the contributions of natural systems to human life. Environmental performance shows how efficiently natural resources are used (Yi et al., 2025).

Q-learning algorithm is used in the creation of balanced decision matrices. For this purpose, the determined decision makers are contacted. In the interview with the decision makers, demographic information and evaluations are collected. The linguistic expressions collected from three different decision makers regarding the comparisons of the criteria with each other are presented in Table A1. Similarly, the same three decision makers evaluate the identified alternatives according to the selected criteria. The linguistic evaluations of the alternatives by the three decision makers according to the criteria are given in Table A2.

The linguistic terms in Table A1 and Table A2 are transformed into MF numbers. Then, Q-learning algorithm is applied to create balanced decision matrices. Since the best decision maker needs to be determined for the Q-learning algorithm, the demographic information of the decision makers is analyzed. Information on age (I1), experience in total (I2), education (I3), profession (I4), published papers (I5), global experience (I6), number of licenses and certificates (I7), teaching experience (I8), number of conference participation (I9) and number of patents (I10) is collected from decision makers. The demographic information about three decision makers is shown in Table A3.

Since each demographic information has different units of measurement, the values are normalized. The normalization process is performed by dividing the values of the information by the maximum value of the information. However, a value is assigned for profession information. A coefficient of 1 is used for professor, 0.9 for board member and 0.8 for chief of executive officer. Similarly, for the education variable, it is defined as 1 since all are PDH. The normalized values obtained as a result are summarized in Table A4. This table includes the average of the normalized values and the weights of the decision makers.

When the weights of the decision makers are examined, it is seen that the best decision maker is the first decision maker. Because the importance weight of Decision Maker 1 is greater than the other decision makers. Therefore, Decision Maker 1 is selected as the best decision maker for the Q-learning algorithm. Firstly, the reward degrees between the best decision maker and the others are calculated with the help of Eq. (4) for the MF relation matrix. The weights of other decision makers are used for the reward factor. The results are displayed in Table A5. Similarly, the reward degrees of the MF decision matrix between the best decision maker and the others are also computed. The reward degrees of MF decision matrix are illustrated in Table A6. Unlike the reward factor, the penalty factor is equal to the weight of the best decision maker. Accordingly, the penalty degrees between the best decision maker and the others are estimated using Eq. (5). The penalty degrees calculated for the MF relation matrix are shared in Table A7. Similarly, the penalty degrees of the MF decision matrix between the best decision maker and the others are also found. The penalty degrees of MF decision matrix are summarized in Table A8.

The values of the relation and decision matrices are updated with Eq. (6). In calculating the updated values, the operations are performed with a learning rate of 0.1. Accordingly, the updated values of the MF relation matrix are reported in Table A9. The same process is performed within the values of the MF decision matrix and the updated values are tabulated in Table A10. The absolute difference is obtained by Eq. (7) and, the result values of Eq. (8) are not less than 0.02. The iteration continues until the result of Eq. (8) is lower than this determined threshold value. At the end of the first iteration, the absolute difference values of the MF relation matrix among the decision makers are shown in Table A11. Since some values of Table A11 are greater than 0.02, the iteration continues. A similar iteration process is performed for the MF decision matrix. The absolute difference values of the MF decision matrix computed as a result of the first iteration are given in Table A12.

Since some values of Table A12 are greater than 0.02, the iteration continues. The results of the next iterations are tabulated in an integrated manner. The iterative results of the molecular relation matrix between Decision Maker 1 and Decision Maker 2 are displayed in Table A13. For the relation matrix, the absolute differences between the first and second decision makers remain below the threshold value at the end of the fifth iteration. However, the absolute difference values between the first and third decision makers are already less than 0.02 at the end of the first iteration. Therefore, iteration is not performed. After the iteration of the relation matrix, the iterations of the decision matrix are performed in a similar manner. The iterative absolute difference of MF decision matrix between first and second decision makers are computed and shown in Table A14. Similar iteration processes are applied between the first and third decision makers. The iterative absolute difference of MF decision matrix between first and third decision makers are estimated and shown in Table A15. For the MF decision matrix, it is seen that the absolute differences between the first and second decision makers become smaller than the threshold value in the fifth iteration, while the absolute differences between the first and third decision makers become smaller than the threshold value in the fourth iteration. Accordingly, the stable MF values for relation matrix are shared in Table A16. Finally, the stable MF values for decision matrix are summarized in Table A17.

4.3. Weighting the criteria for environmental impact assessment using MF-BANEW

The average of the MF evaluations in Table A16 is calculated with the help of Eq. (9) and the averaged MF relation matrix is constructed. The averaged MF relation matrix is shared in Table A18. Afterwards, the fuzzy vectors defined in Eq. (10) are created. For relation matrix, fuzzy vectors are summarized in Table A19. Angles in radians for the relation matrix are calculated using fuzzy vectors using Eq. (11). Angles in radians for the relation matrix are presented in Table A20. While the maximum value is usually used in the normalization of angles as in Eq. (12), in this article the values are normalized by applying the operations in Eq. (13). Normalized values corresponding to different molecular geometry shapes are presented in Table A21. The reciprocal values of the normalized angle values calculated for each molecular geometric shape are determined by Eq. (14). The reciprocal values computed for the linear shape are given in Table A22. The reciprocal values of the normalized angle values for the linear shape are normalized with Eq. (15) and the matrix formed in Eq. (16) is created. The normalized relation matrix for linear shape is displayed in Table A23. Finally, entropy values are determined with the help of Eqs. (17) and (18) and information gain values are estimated using Eq. (19). Eq. (20) presents the criterion weights by these values. All results for linear shape are summarized in Table 1.

According to Table 1, the most important criterion for environmental impact assessment is biodiversity. For consistency and validity of the criteria's order of priority, results for different learning rates and other molecular geometry shapes are compared. The comparison

Table 1
The results of entropy, information gain and weights for linear shape.

	$E(C_i)$	$E(C_i\|C_j)$	$Gain(C_i)$	$w(C_i)$
BIODIV	1.386	0.755	0.631	0.273
WTRRES		0.854	0.532	0.230
AIRQUA		0.825	0.562	0.243
LNDPRD		0.801	0.585	0.253

prioritization results are visualized in Fig. 3.

It can be seen in Fig. 3 that the prioritization results based on different molecular geometry shapes and learning rates are similar. According to the results of Fig. 3, the prioritization of the criteria is the same and the analysis is valid and consistent.

4.4. Ranking the alternative strategies for assessing environmental impact in renewable energy investments using MF-MOPSO

The average of the MF evaluations in Table A17 is computed with the help of Eq. (9) and the averaged MF decision matrix is created. The averaged MF relation matrix is illustrated in Table A24. Then, the weighted MF decision matrix is calculated using the criterion weights obtained with MF-BANEW with the help of Eq. (21). The weighted MF decision matrix is shown in Table A25. The fuzzy vectors from the elements of the weighted MF decision matrix are defined by Eq. (22). MF vectors for decision matrix are shared in Table A26. Angles in radians for the decision matrix are computed with the help of fuzzy vectors using Eq. (23). Angles in radians for the decision matrix are summarized in Table A27. While the maximum value is generally used in the normalization of angles as in Eq. (24), in this article the angle values for the MF decision matrix are normalized by applying the operations in Eq. (25). Normalized values corresponding to different molecular geometry shapes are displayed in Table A28. The reciprocal values of the normalized angle values calculated for each molecular geometric shape are determined by Eq. (26). The reciprocal values computed for the linear shape are given in Table A29. The reciprocal values of the normalized angle values for the linear shape are normalized with Eq. (27) and the final decision matrix is constructed. The normalized decision matrix for linear shape is illustrated in Table A30. After obtaining the final decision matrix, MOPSO algorithm is used. Eqs. (28)–(34) are applied iteratively. The iteration results are tabulated in Table A31.

As a result of the fourth iteration, since the change in global best positions is less than the threshold value of 0.001, the values found as a result of the fourth iteration are accepted as the final values. The average of the final values is estimated by Eq. (35). The average values of the alternatives strategies for assessing environmental impact in renewable energy investments are calculated as 0.1398, 0.1354, 0.1153, 0.1233, and 0.1135 respectively. So, the ranking results are given in descending order as LFCYAS, DYSYSOT, ECOSRV, GEOINFSYS, ENVPER. In other words, the most optimal alternative strategy for assessing environmental impact in renewable energy investments is life cycle assessment. For consistency and validity of the ranking, results for different learning rates and other molecular geometry shapes are compared. The comparison ranking results are visualized in Fig. 4.

It can be seen in Fig. 4 that the ranking results based on different molecular geometry shapes and learning rates are similar. According to the results of Fig. 4, the ranking of the alternatives is the same and the analysis is valid and consistent.

5. Discussion

The results of the analysis show that the environmental impact of renewable energy investments is mostly on biodiversity. Renewable energy investments are an ideal alternative for the protection of the natural environment. Moreover, it is important for biodiversity as it reduces carbon emissions. In addition, renewable energy has a significant impact on combating climate change. Finally, the harmful chemicals and waste generated during the extraction of fossil fuels are not encountered in the renewable energy process. Joof et al. (2023) examined the impact of the insurance market, climate change, and renewable energy on biodiversity in BRICS economies using the nonlinear

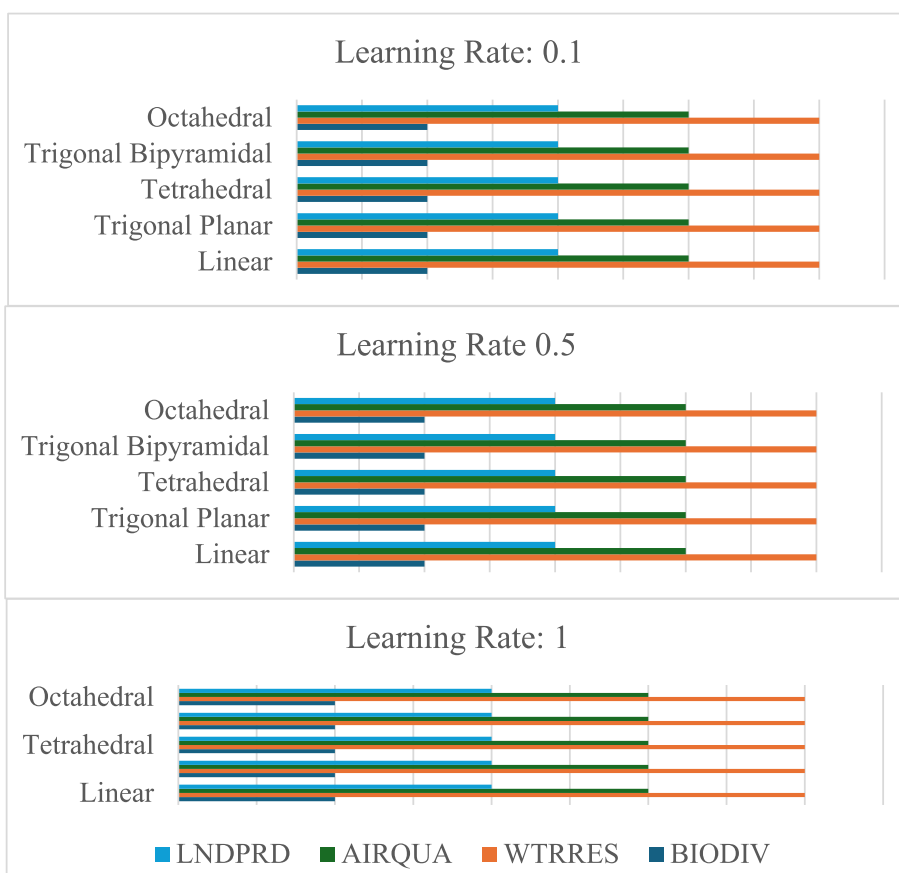

Fig. 3. The comparison prioritization results of the criteria.

Fig. 4. The comparison ranking results of the alternatives.

autoregressive distributed lag method. It is argued that increasing the use of renewable energy reduces biodiversity loss. Guo et al. (2023) analyzed the relationship between organic agriculture, renewable energy, and biodiversity. The results of the study conducted in Europe advocate that the use of renewable energy increases the biodiversity crisis.

The assessment results show that land productivity is the second most important criterion. Renewable energy does not cause soil contamination like fossil fuels. In this way, land productivity is increased. Among renewable energy types, especially biomass energy allows organic wastes to be used as fertilizer. Organic fertilizer contributes to increased land productivity. Raihan (2023) studied the relationship between renewable energy use and agricultural productivity in the Philippines. It is argued that the use of renewable energy should be expanded to ensure environmental sustainability. Shah et al. (2023) examined the role of renewable energy, environmental policies, and technology in agricultural productivity. The results of the research conducted in BRICS countries show that the use of renewable energy increases agricultural productivity.

It is identified that the most optimal option for assessing the environmental impact of renewable energy investments is life cycle assessment. This method addresses the environmental impacts of renewable energy investments with a holistic approach. It is an approach that performs environmental impact assessment not only during the utilization process but also by taking into consideration all other processes such as production and maintenance. Kaczmarczyk et al. (2024) conducted a life cycle assessment of a distillation system powered by renewable energy. It is reported that the use of distillation with renewable energy is a promising development. The results of the analysis indicate that the second most ideal alternative is dynamic systems over time. The energy production, consumption, storage, and political factors of renewable energy investments are constantly changing. Therefore, the evaluation of these investments requires the analysis of dynamic systems. Coppitters and Contino (2023) carried out a study to prevent vulnerability in renewable energy systems. It is underlined that renewable energy processes are dynamic.

6. Conclusion

6.1. Concluding remarks

This study focuses on significant investment strategies to improve the environmental impact of the renewable energy projects. For this purpose, a novel model is constructed that has 3 stages and 26 steps. At the first stage, the balanced expert dataset has been created by Q-learning algorithm. The following stage evaluates the criteria with MF BANEW-based weighting. In the third stage, alternatives are ranked by MF MOPSO. It is concluded that biodiversity is the most critical environmental impact of renewable energy investments is mostly on biodiversity. It is also defined that land productivity is the second most important criterion in this regard. Furthermore, the ranking results denote that the most optimal option for assessing the environmental impact of renewable energy investments is life cycle assessment. The main contribution of this study to the literature is the determination of priority investment strategies with a new model for the development of more environmentally sensitive renewable energy projects. The use of molecular geometry and fuzzy multi-criteria decision-making analysis together increases the quality of the proposed model. This situation reduces the uncertainty in the solution process of complex problems more effectively. Moreover, the consideration of the Q-learning algorithm in the expert weighting process has a positive contribution to the decision-making literature.

6.2. Policy implications

Considering the findings of this study, some policy recommendations can be determined to reduce the environmental impacts of renewable energy projects. First, investment incentives can be applied for the protection of biodiversity. According to the findings, biodiversity is the most critical environmental impact of renewable energy investments. Therefore, policy makers should provide incentives for projects that protect biodiversity. The fact that corporate tax is not collected from businesses making these investments allows these investors to focus on this area. The cost advantage provided as a result of the tax exemption increases the interest of investors in these projects. Similarly, legal regulations should be established to prevent renewable energy projects from being established in regions sensitive to biodiversity. This situation contributes significantly to minimizing the environmental damage of the projects. The effective implementation of these regulations is also of vital importance for the effectiveness of this process. In this context, periodic inspections should be carried out to ensure the control of these practices. In case of non-compliance with these rules, high financial sanctions should be applied. This situation has a significant effect on businesses not establishing projects in regions sensitive to biodiversity. Similarly, financial incentives should be provided for projects with high environmental sensitivity. For this purpose, low-interest loans should be provided. This situation provides significant cost advantages to the projects. On the other hand, land productivity has been determined as

the second most important criterion in renewable energy projects. In this context, renewable energy projects should be established on lands that do not have agricultural value. In this way, it is possible to significantly reduce the damage caused by renewable energy projects to agricultural production. For this process to take place correctly, the necessary legal regulations should be established. For these regulations to be defined correctly, it is necessary to clearly determine where the agricultural lands are located within the country. In addition to them, the study results show that the most optimal way to assess environmental impact is life cycle assessment. To achieve this goal, this situation should be made a legal obligation. Especially in large-scale projects, life cycle assessment results should be a part of the project approval process.

6.3. Limitations and future research directions

Renewable energy projects can show great diversity in terms of geographical conditions, social and economic factors. In this context, an analysis can be carried out for specific country groups in future studies. Although an innovative model is presented in this study, there are also some limitations in this model. A limited number of expert opinions are considered in the study. In this context, the experiences and perspectives of the experts used may not cover a broad perspective. In this context, more expert opinions can be taken into consideration in a new model to be created in future studies. On the other hand, finding experts with sufficient work experience for renewable energy investments is a very difficult issue. Because of this situation, virtual experts can be created using artificial intelligence or machine learning techniques. This allows for more criteria and alternative evaluations to be reached. Similarly, the structure of the model, consisting of 3 stages and 26 steps, may create some difficulties in the application and analysis processes. As a future research direction, a model that is easy to apply in practice can be developed. In parallel, some improvements can be made to this model and its complexity can be reduced.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
The linguistic expressions regarding the comparisons of the criteria.

Decision maker 1	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV		M	M	S
WTRRES	S		H	S
AIRQUA	H	H		S
LNDPRD	S	M	S	
Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV		M	M	M
WTRRES	S		M	M
AIRQUA	M	H		S
LNDPRD	S	M	S	
Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV		S	S	S
WTRRES	S		H	S
AIRQUA	S	H		S
LNDPRD	S	M	S	

Table A2
The linguistic evaluations of the alternatives according to criteria.

Decision maker 1	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	M	H	M	M
DYSYSOT	M	M	H	M
GEOINFSYS	H	S	S	S
ECOSRV	S	M	M	H
ENVPER	S	H	H	H
Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	H	H	S	M
DYSYSOT	S	S	H	S
GEOINFSYS	H	S	S	H
ECOSRV	H	S	S	H
ENVPER	S	H	H	H
Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	S	H	M	M
DYSYSOT	S	S	H	H
GEOINFSYS	H	H	H	S
ECOSRV	H	S	S	H
ENVPER	S	H	H	H

Table A3
The demographic information about three decision makers.

	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10
Decision Maker 1	50	25	PhD	Professor	95	12	4	23	40	2
Decision Maker 2	45	21	PhD	Board Member	12	7	5	3	14	4
Decision Maker 3	47	18	PhD	Chief of Executive Officer	7	11	4	2	8	3

Table A4
The normalized demographic information and weights.

	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10	Average	Weights
Decision Maker 1	1	1	1	1	1	1	0.80	1	1	0.50	0.93	0.415
Decision Maker 2	0.90	0.84	1	0.80	0.13	0.58	1	0.13	0.35	1	0.67	0.300
Decision Maker 3	0.94	0.72	1	0.90	0.07	0.92	0.80	0.09	0.20	0.75	0.64	0.285

Table A5
Reward degrees of MF relation matrix.

Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(−0.06, 0.05, 0.02)
WTRRES	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(−0.11, 0.08, 0.03)	(−0.06, 0.05, 0.02)
AIRQUA	(−0.11, 0.08, 0.03)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
LNDPRD	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.06, −0.04, −0.01)	(0.06, −0.04, −0.01)	(0.00, 0.00, 0.00)
WTRRES	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
AIRQUA	(−0.04, 0.03, 0.01)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
LNDPRD	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)

Table A6
The reward degrees of MF decision matrix.

Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.11, −0.08, −0.03)	(0.00, 0.00, 0.00)	(0.06, −0.05, −0.02)	(0.00, 0.00, 0.00)
DYSYSOT	(0.06, −0.05, −0.02)	(0.06, −0.05, −0.02)	(0.00, 0.00, 0.00)	(0.06, −0.05, −0.02)
GEOINFSYS	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.05, −0.03, −0.02)
ECOSRV	(0.05, −0.03, −0.02)	(0.06, −0.05, −0.02)	(0.06, −0.05, −0.02)	(0.00, 0.00, 0.00)
ENVPER	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.06, −0.04, −0.01)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
DYSYSOT	(0.06, −0.04, −0.01)	(0.06, −0.04, −0.01)	(0.00, 0.00, 0.00)	(0.10, −0.07, −0.03)
GEOINFSYS	(0.00, 0.00, 0.00)	(0.04, −0.03, −0.01)	(0.04, −0.03, −0.01)	(0.00, 0.00, 0.00)
ECOSRV	(0.04, −0.03, −0.01)	(0.06, −0.04, −0.01)	(0.06, −0.04, −0.01)	(0.00, 0.00, 0.00)
ENVPER	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)

Table A7
Penalty degrees of MF relation matrix.

Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.08, −0.06, −0.02)
WTRRES	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.15, −0.10, −0.04)	(0.08, −0.06, −0.02)
AIRQUA	(0.15, −0.10, −0.04)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
LNDPRD	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(−0.08, 0.06, 0.02)	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)
WTRRES	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
AIRQUA	(0.06, −0.04, −0.02)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
LNDPRD	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)

Table A8
The penalty degrees of MF decision matrix.

Decision maker 1-Decision maker 2	BIODIV	WTRRES	AIRQUA	LNDPRD
LFCYAS	(−0.15, 0.10, 0.04)	(0.00, 0.00, 0.00)	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)
DYSYSOT	(−0.08, 0.06, 0.02)	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)	(−0.08, 0.06, 0.02)
GEOINFSYS	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(−0.06, 0.04, 0.02)
ECOSRV	(−0.06, 0.04, 0.02)	(−0.08, 0.06, 0.02)	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)
ENVPER	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)
DYSYSOT	(−0.08, 0.06, 0.02)	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)	(−0.15, 0.10, 0.04)
GEOINFSYS	(0.00, 0.00, 0.00)	(−0.06, 0.04, 0.02)	(−0.06, 0.04, 0.02)	(0.00, 0.00, 0.00)
ECOSRV	(−0.06, 0.04, 0.02)	(−0.08, 0.06, 0.02)	(−0.08, 0.06, 0.02)	(0.00, 0.00, 0.00)
ENVPER	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)	(0.00, 0.00, 0.00)

Table A9
The updated values of the MF relation matrix.

Decision maker 1-Decision maker 2	BIODIV	WTRRES	AIRQUA	LNDPRD
BIODIV	(0.00, 0.00, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)	(0.79, 0.16, 0.05)
WTRRES	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)	(0.92, 0.07, 0.01)	(0.79, 0.16, 0.05)
AIRQUA	(0.92, 0.07, 0.01)	(0.95, 0.05, 0.00)	(0.00, 0.00, 0.00)	(0.80, 0.15, 0.05)
LNDPRD	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.80, 0.15, 0.05)
WTRRES	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)	(0.95, 0.05, 0.00)	(0.80, 0.15, 0.05)
AIRQUA	(0.94, 0.06, 0.00)	(0.95, 0.05, 0.00)	(0.00, 0.00, 0.00)	(0.80, 0.15, 0.05)
LNDPRD	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)

Table A10
The updated values of the MF decision matrix.

Decision maker 1-Decision maker 2	BIODIV	WTRRES	AIRQUA	LNDPRD
LFCYAS	(0.63, 0.28, 0.09)	(0.95, 0.05, 0.00)	(0.61, 0.29, 0.10)	(0.60, 0.30, 0.10)
DYSYSOT	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)	(0.61, 0.29, 0.10)
GEOINFSYS	(0.95, 0.05, 0.00)	(0.80, 0.15, 0.05)	(0.80, 0.15, 0.05)	(0.81, 0.14, 0.05)
ECOSRV	(0.81, 0.14, 0.05)	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)
ENVPER	(0.80, 0.15, 0.05)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)
DYSYSOT	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)	(0.62, 0.28, 0.09)
GEOINFSYS	(0.95, 0.05, 0.00)	(0.81, 0.14, 0.05)	(0.81, 0.14, 0.05)	(0.80, 0.15, 0.05)
ECOSRV	(0.81, 0.14, 0.05)	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)
ENVPER	(0.80, 0.15, 0.05)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)

Table A11
The absolute difference values of the MF relation matrix (First iteration).

Decision maker 1-Decision maker 2	BIODIV	WTRRES	AIRQUA	LNDPRD
BIODIV	(0.000, 0.000, 0.000)	(0.000, 0.000, 0.000)	(0.000, 0.000, 0.000)	(0.014, 0.11, 0.04)
WTRRES	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.025, 0.18, 0.07)	(0.014, 0.11, 0.04)
AIRQUA	(0.025, 0.18, 0.07)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
LNDPRD	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.000, 0.00, 0.00)	(0.014, 0.10, 0.03)	(0.014, 0.10, 0.03)	(0.000, 0.00, 0.00)
WTRRES	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
AIRQUA	(0.010, 0.07, 0.03)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
LNDPRD	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)

Table A12
The absolute difference values of the MF decision matrix (First iteration).

Decision maker 1-Decision maker 2	BIODIV	WTRRES	AIRQUA	LNDPRD
LFCYAS	(0.025, 0.18, 0.07)	(0.000, 0.00, 0.00)	(0.014, 0.11, 0.04)	(0.000, 0.00, 0.00)
DYSYSOT	(0.014, 0.11, 0.04)	(0.014, 0.11, 0.04)	(0.000, 0.00, 0.00)	(0.014, 0.11, 0.04)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.011, 0.07, 0.04)
ECOSRV	(0.011, 0.07, 0.04)	(0.014, 0.11, 0.04)	(0.014, 0.11, 0.04)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.014, 0.10, 0.03)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
DYSYSOT	(0.014, 0.10, 0.03)	(0.014, 0.10, 0.03)	(0.000, 0.00, 0.00)	(0.024, 0.17, 0.07)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.010, 0.07, 0.03)	(0.010, 0.07, 0.03)	(0.000, 0.00, 0.00)
ECOSRV	(0.010, 0.07, 0.03)	(0.014, 0.10, 0.03)	(0.014, 0.10, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)

Table A13
The iterative results of the molecular relation matrix between first and second decision makers.

Iteration 2				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.013, 0.10, 0.03)
WTRRES	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.023, 0.17, 0.07)	(0.013, 0.10, 0.03)
AIRQUA	(0.023, 0.17, 0.07)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
LNDPRD	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 3				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.012, 0.09, 0.03)
WTRRES	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.022, 0.15, 0.06)	(0.012, 0.09, 0.03)
AIRQUA	(0.022, 0.15, 0.06)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
LNDPRD	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 4				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.011, 0.09, 0.03)
WTRRES	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.020, 0.14, 0.06)	(0.011, 0.09, 0.03)
AIRQUA	(0.020, 0.14, 0.06)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
LNDPRD	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 5				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.011, 0.08, 0.03)
WTRRES	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.019, 0.13, 0.05)	(0.011, 0.08, 0.03)
AIRQUA	(0.019, 0.13, 0.05)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
LNDPRD	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)

Table A14
The iterative results of the molecular decision matrix between first and second decision makers.

Iteration 2				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.023, 0.17, 0.07)	(0.000, 0.00, 0.00)	(0.013, 0.10, 0.03)	(0.000, 0.00, 0.00)
DYSYSOT	(0.013, 0.10, 0.03)	(0.013, 0.10, 0.03)	(0.000, 0.00, 0.00)	(0.013, 0.10, 0.03)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.010, 0.07, 0.03)
ECOSRV	(0.010, 0.07, 0.03)	(0.013, 0.10, 0.03)	(0.013, 0.10, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 3				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.022, 0.15, 0.06)	(0.000, 0.00, 0.00)	(0.012, 0.09, 0.03)	(0.000, 0.00, 0.00)
DYSYSOT	(0.012, 0.09, 0.03)	(0.012, 0.09, 0.03)	(0.000, 0.00, 0.00)	(0.012, 0.09, 0.03)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.009, 0.06, 0.03)
ECOSRV	(0.009, 0.06, 0.03)	(0.012, 0.09, 0.03)	(0.012, 0.09, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 4				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.020, 0.14, 0.06)	(0.000, 0.00, 0.00)	(0.011, 0.09, 0.03)	(0.000, 0.00, 0.00)
DYSYSOT	(0.011, 0.09, 0.03)	(0.011, 0.09, 0.03)	(0.000, 0.00, 0.00)	(0.011, 0.09, 0.03)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.009, 0.06, 0.03)
ECOSRV	(0.009, 0.06, 0.03)	(0.011, 0.09, 0.03)	(0.011, 0.09, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 5				
Decision maker 1-Decision maker 2	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.019, 0.13, 0.05)	(0.000, 0.00, 0.00)	(0.011, 0.08, 0.03)	(0.000, 0.00, 0.00)
DYSYSOT	(0.011, 0.08, 0.03)	(0.011, 0.08, 0.03)	(0.000, 0.00, 0.00)	(0.011, 0.08, 0.03)

(*continued on next page*)

Table A14 (*continued*)

Iteration 2 **Decision maker 1-Decision maker 2**	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
GEOINFSYS	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.008, 0.05, 0.03)
ECOSRV	(0.008, 0.05, 0.03)	(0.011, 0.08, 0.03)	(0.011, 0.08, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)

Table A15
The iterative results of the molecular decision matrix between first and third decision makers.

Iteration 2 **Decision maker 1-Decision maker 3**	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.013, 0.10, 0.03)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
DYSYSOT	(0.013, 0.10, 0.03)	(0.013, 0.10, 0.03)	(0.000, 0.00, 0.00)	(0.023, 0.16, 0.07)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.010, 0.07, 0.03)	(0.010, 0.07, 0.03)	(0.000, 0.00, 0.00)
ECOSRV	(0.010, 0.07, 0.03)	(0.013, 0.10, 0.03)	(0.013, 0.10, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 3				
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.012, 0.09, 0.03)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
DYSYSOT	(0.012, 0.09, 0.03)	(0.012, 0.09, 0.03)	(0.000, 0.00, 0.00)	(0.021, 0.15, 0.06)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.009, 0.06, 0.03)	(0.009, 0.06, 0.03)	(0.000, 0.00, 0.00)
ECOSRV	(0.009, 0.06, 0.03)	(0.012, 0.09, 0.03)	(0.012, 0.09, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
Iteration 4				
Decision maker 1-Decision maker 3	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.011, 0.08, 0.03)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)
DYSYSOT	(0.011, 0.08, 0.03)	(0.011, 0.08, 0.03)	(0.000, 0.00, 0.00)	(0.020, 0.14, 0.06)
GEOINFSYS	(0.000, 0.00, 0.00)	(0.008, 0.06, 0.03)	(0.008, 0.06, 0.03)	(0.000, 0.00, 0.00)
ECOSRV	(0.008, 0.06, 0.03)	(0.011, 0.08, 0.03)	(0.011, 0.08, 0.03)	(0.000, 0.00, 0.00)
ENVPER	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)	(0.000, 0.00, 0.00)

Table A16
Balanced MF evaluations for relation matrix.

Decision maker 1(initial Q matrix)	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)
WTRRES	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)	(0.95, 0.05, 0.00)	(0.80, 0.15, 0.05)
AIRQUA	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.00, 0.00, 0.00)	(0.80, 0.15, 0.05)
LNDPRD	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)
Decision maker 2 (Balanced Q matrix)	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)	(0.74, 0.20, 0.07)
WTRRES	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)	(0.84, 0.13, 0.03)	(0.74, 0.20, 0.07)
AIRQUA	(0.84, 0.13, 0.03)	(0.95, 0.05, 0.00)	(0.00, 0.00, 0.00)	(0.80, 0.15, 0.05)
LNDPRD	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)
Decision maker 3 (Balanced Q matrix)	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV	(0.00, 0.00, 0.00)	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.80, 0.15, 0.05)
WTRRES	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)	(0.95, 0.05, 0.00)	(0.80, 0.15, 0.05)
AIRQUA	(0.94, 0.06, 0.00)	(0.95, 0.05, 0.00)	(0.00, 0.00, 0.00)	(0.80, 0.15, 0.05)
LNDPRD	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)

Table A17
Balanced MF evaluations for decision matrix.

Decision maker 1(initial Q matrix)	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.60, 0.30, 0.10)	(0.95, 0.05, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)
DYSYSOT	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)	(0.95, 0.05, 0.00)	(0.60, 0.30, 0.10)
GEOINFSYS	(0.95, 0.05, 0.00)	(0.80, 0.15, 0.05)	(0.80, 0.15, 0.05)	(0.80, 0.15, 0.05)
ECOSRV	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)	(0.95, 0.05, 0.00)
ENVPER	(0.80, 0.15, 0.05)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)
Decision maker 2 (Balanced Q matrix)	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.63, 0.28, 0.09)	(0.95, 0.05, 0.00)	(0.61, 0.29, 0.10)	(0.60, 0.30, 0.10)
DYSYSOT	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)	(0.61, 0.29, 0.10)
GEOINFSYS	(0.95, 0.05, 0.00)	(0.80, 0.15, 0.05)	(0.80, 0.15, 0.05)	(0.81, 0.14, 0.05)
ECOSRV	(0.81, 0.14, 0.05)	(0.61, 0.29, 0.10)	(0.61, 0.29, 0.10)	(0.95, 0.05, 0.00)
ENVPER	(0.80, 0.15, 0.05)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)
Decision maker 3 (Balanced Q matrix)	**BIODIV**	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.65, 0.26, 0.09)	(0.95, 0.05, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)
DYSYSOT	(0.65, 0.26, 0.09)	(0.65, 0.26, 0.09)	(0.95, 0.05, 0.00)	(0.69, 0.24, 0.07)

(*continued on next page*)

Table A17 (*continued*)

Decision maker 1(initial Q matrix)	BIODIV	WTRRES	AIRQUA	LNDPRD
GEOINFSYS	(0.95, 0.05, 0.00)	(0.84, 0.12, 0.04)	(0.84, 0.12, 0.04)	(0.80, 0.15, 0.05)
ECOSRV	(0.84, 0.12, 0.04)	(0.65, 0.26, 0.09)	(0.65, 0.26, 0.09)	(0.95, 0.05, 0.00)
ENVPER	(0.80, 0.15, 0.05)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)

Table A18
The averaged MF relation matrix.

	BIODIV	WTRRES	AIRQUA	LNDPRD
BIODIV	(0.00, 0.00, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)	(0.78, 0.17, 0.06)
WTRRES	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)	(0.91, 0.08, 0.01)	(0.78, 0.17, 0.06)
AIRQUA	(0.91, 0.08, 0.01)	(0.95, 0.05, 0.00)	(0.00, 0.00, 0.00)	(0.80, 0.15, 0.05)
LNDPRD	(0.80, 0.15, 0.05)	(0.60, 0.30, 0.10)	(0.80, 0.15, 0.05)	(0.00, 0.00, 0.00)

Table A19
The vectors for MF relation matrix.

	Vectors
$\mathscr{L}_1$	[(0.60, 0.30, 0.10), (0.60, 0.30, 0.10), (0.78, 0.17, 0.06)]
$\mathscr{L}_2$	[(0.80, 0.15, 0.05), (0.91, 0.08, 0.01), (0.78, 0.17, 0.06)]
$\mathscr{L}_3$	[(0.91, 0.08, 0.01), (0.95, 0.05, 0.00), (0.80, 0.15, 0.05)]
$\mathscr{L}_4$	[(0.80, 0.15, 0.05), (0.60, 0.30, 0.10), (0.80, 0.15, 0.05)]

Table A20
The angles in radians for the MF relation matrix.

	$\theta_{\mathscr{L}_1}$	$\theta_{\mathscr{L}_2}$	$\theta_{\mathscr{L}_3}$	$\theta_{\mathscr{L}_4}$
$\theta_{\mathscr{L}_1}$		0.305	0.360	0.182
$\theta_{\mathscr{L}_2}$	0.305		0.079	0.270
$\theta_{\mathscr{L}_3}$	0.360	0.079		0.286
$\theta_{\mathscr{L}_4}$	0.182	0.270	0.286	

Table A21
The normalized angle values.

Linear	$\aleph(\theta_{\mathscr{L}_1})$	$\aleph(\theta_{\mathscr{L}_2})$	$\aleph(\theta_3)$	$\aleph(\theta_{\mathscr{L}_4})$
$\aleph(\theta_{\mathscr{L}_1})$	0.000	0.097	0.115	0.058
$\aleph(\theta_{\mathscr{L}_2})$	0.097	0.000	0.025	0.086
$\aleph(\theta_3)$	0.115	0.025	0.000	0.091
$\aleph(\theta_{\mathscr{L}_4})$	0.058	0.086	0.091	0.000
Trigonal Planar	$\aleph(\theta_{\mathscr{L}_1})$	$\aleph(\theta_{\mathscr{L}_2})$	$\aleph(\theta_3)$	$\aleph(\theta_{\mathscr{L}_4})$
$\aleph(\theta_{\mathscr{L}_1})$	0.000	0.145	0.172	0.087
$\aleph(\theta_{\mathscr{L}_2})$	0.145	0.000	0.038	0.129
$\aleph(\theta_3)$	0.172	0.038	0.000	0.137
$\aleph(\theta_{\mathscr{L}_4})$	0.087	0.129	0.137	0.000
Tetrahedral	$\aleph(\theta_{\mathscr{L}_1})$	$\aleph(\theta_{\mathscr{L}_2})$	$\aleph(\theta_3)$	$\aleph(\theta_{\mathscr{L}_4})$
$\aleph(\theta_{\mathscr{L}_1})$	0.000	0.159	0.189	0.095
$\aleph(\theta_{\mathscr{L}_2})$	0.159	0.000	0.041	0.141
$\aleph(\theta_3)$	0.189	0.041	0.000	0.150
$\aleph(\theta_{\mathscr{L}_4})$	0.095	0.141	0.150	0.000
Trigonal Bipyramidal	$\aleph(\theta_{\mathscr{L}_1})$	$\aleph(\theta_{\mathscr{L}_2})$	$\aleph(\theta_3)$	$\aleph(\theta_{\mathscr{L}_4})$
$\aleph(\theta_{\mathscr{L}_1})$	0.000	0.166	0.197	0.099
$\aleph(\theta_{\mathscr{L}_2})$	0.166	0.000	0.043	0.147
$\aleph(\theta_3)$	0.197	0.043	0.000	0.156
$\aleph(\theta_{\mathscr{L}_4})$	0.099	0.147	0.156	0.000
Octahedral	$\aleph(\theta_{\mathscr{L}_1})$	$\aleph(\theta_{\mathscr{L}_2})$	$\aleph(\theta_3)$	$\aleph(\theta_{\mathscr{L}_4})$
$\aleph(\theta_{\mathscr{L}_1})$	0.000	0.194	0.229	0.116
$\aleph(\theta_{\mathscr{L}_2})$	0.194	0.000	0.050	0.172
$\aleph(\theta_3)$	0.229	0.050	0.000	0.182
$\aleph(\theta_{\mathscr{L}_4})$	0.116	0.172	0.182	0.000

Table A22
The reciprocal values for linear shape.

	$\Im(e_{\mathscr{L}_1})$	$\Im(e_{\mathscr{L}_2})$	$\Im(e_{\mathscr{L}_3})$	$\Im(e_{\mathscr{L}_4})$
$\Im(e_{\mathscr{L}_1})$		10.314	8.717	17.283
$\Im(e_{\mathscr{L}_2})$	10.314		39.873	11.652
$\Im(e_{\mathscr{L}_3})$	8.717	39.873		10.987
$\Im(e_{\mathscr{L}_4})$	17.283	11.652	10.987	

Table A23
The normalized relation matrix for linear shape.

	BIODIV	**WTRRES**	**AIRQUA**	**LNDPRD**
BIODIV		0.104	0.088	0.175
WTRRES	0.104		0.403	0.118
AIRQUA	0.088	0.403		0.111
LNDPRD	0.175	0.118	0.111	

Table A24
The averaged MF decision matrix.

	BIODIV	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.63, 0.28, 0.09)	(0.95, 0.05, 0.00)	(0.60, 0.30, 0.10)	(0.60, 0.30, 0.10)
DYSYSOT	(0.62, 0.28, 0.09)	(0.62, 0.28, 0.09)	(0.95, 0.05, 0.00)	(0.63, 0.28, 0.09)
GEOINFSYS	(0.95, 0.05, 0.00)	(0.81, 0.14, 0.05)	(0.81, 0.14, 0.05)	(0.80, 0.15, 0.05)
ECOSRV	(0.82, 0.14, 0.04)	(0.62, 0.28, 0.09)	(0.62, 0.28, 0.09)	(0.95, 0.05, 0.00)
ENVPER	(0.80, 0.15, 0.05)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)	(0.95, 0.05, 0.00)

Table A25
The weighted MF decision matrix.

	BIODIV	**WTRRES**	**AIRQUA**	**LNDPRD**
LFCYAS	(0.17, 0.08, 0.03)	(0.22, 0.01, 0.00)	(0.15, 0.07, 0.02)	(0.15, 0.08, 0.03)
DYSYSOT	(0.17, 0.08, 0.03)	(0.14, 0.07, 0.02)	(0.23, 0.01, 0.00)	(0.16, 0.07, 0.02)
GEOINFSYS	(0.26, 0.01, 0.00)	(0.19, 0.03, 0.01)	(0.20, 0.03, 0.01)	(0.20, 0.04, 0.01)
ECOSRV	(0.22, 0.04, 0.01)	(0.14, 0.07, 0.02)	(0.15, 0.07, 0.02)	(0.24, 0.01, 0.00)
ENVPER	(0.22, 0.04, 0.01)	(0.22, 0.01, 0.00)	(0.23, 0.01, 0.00)	(0.24, 0.01, 0.00)

Table A26
The vectors for MF decision matrix.

	Vectors
Ω_1	[(0.17, 0.08, 0.03), (0.22, 0.01, 0.00), (0.15, 0.07, 0.02), (0.15, 0.08, 0.03)]
Ω_2	[(0.17, 0.08, 0.03), (0.14, 0.07, 0.02), (0.23, 0.01, 0.00), (0.16, 0.07, 0.02)]
Ω_3	[(0.26, 0.01, 0.00), (0.19, 0.03, 0.01), (0.20, 0.03, 0.01), (0.20, 0.04, 0.01)]
Ω_4	[(0.22, 0.04, 0.01), (0.14, 0.07, 0.02), (0.15, 0.07, 0.02), (0.24, 0.01, 0.00)]
Ω_5	[(0.22, 0.04, 0.01), (0.22, 0.01, 0.00), (0.23, 0.01, 0.00), (0.24, 0.01, 0.00)]

Table A27
The angles in radians for the MF decision matrix.

	$\mathscr{I}_{\Omega_1}$	$\mathscr{I}_{\Omega_2}$	$\mathscr{I}_{\Omega_3}$	$\mathscr{I}_{\Omega_4}$	$\mathscr{I}_{\Omega_5}$
$\mathscr{I}_{\Omega_1}$		0.380	0.346	0.411	0.349
$\mathscr{I}_{\Omega_2}$	0.380		0.332	0.401	0.317
$\mathscr{I}_{\Omega_3}$	0.346	0.332		0.240	0.194
$\mathscr{I}_{\Omega_4}$	0.411	0.401	0.240		0.299
$\mathscr{I}_{\Omega_5}$	0.349	0.317	0.194	0.299	

Table A28
The normalized angle values.

Linear	$\aleph(\mathscr{g}_{\Omega_1})$	$\aleph(\mathscr{g}_{\Omega_2})$	$\aleph(\mathscr{g}_{\Omega_3})$	$\aleph(\mathscr{g}_{\Omega_4})$	$\aleph(\mathscr{g}_{\Omega_5})$
$\aleph(\mathscr{g}_{\Omega_1})$	0.000	0.121	0.110	0.131	0.111
$\aleph(\mathscr{g}_{\Omega_2})$	0.121	0.000	0.106	0.128	0.101
$\aleph(\mathscr{g}_{\Omega_3})$	0.110	0.106	0.000	0.076	0.062
$\aleph(\mathscr{g}_{\Omega_4})$	0.131	0.128	0.076	0.000	0.095
$\aleph(\mathscr{g}_{\Omega_5})$	0.111	0.101	0.062	0.095	0.000
Trigonal Planar	$\aleph(\mathscr{g}_{\Omega_1})$	$\aleph(\mathscr{g}_{\Omega_2})$	$\aleph(\mathscr{g}_{\Omega_3})$	$\aleph(\mathscr{g}_{\Omega_4})$	$\aleph(\mathscr{g}_{\Omega_5})$
$\aleph(\mathscr{g}_{\Omega_1})$	0.000	0.181	0.165	0.196	0.167
$\aleph(\mathscr{g}_{\Omega_2})$	0.181	0.000	0.159	0.191	0.151
$\aleph(\mathscr{g}_{\Omega_3})$	0.165	0.159	0.000	0.115	0.093
$\aleph(\mathscr{g}_{\Omega_4})$	0.196	0.191	0.115	0.000	0.143
$\aleph(\mathscr{g}_{\Omega_5})$	0.167	0.151	0.093	0.143	0.000
Tetrahedral	$\aleph(\mathscr{g}_{\Omega_1})$	$\aleph(\mathscr{g}_{\Omega_2})$	$\aleph(\mathscr{g}_{\Omega_3})$	$\aleph(\mathscr{g}_{\Omega_4})$	$\aleph(\mathscr{g}_{\Omega_5})$
$\aleph(\mathscr{g}_{\Omega_1})$	0.000	0.199	0.181	0.215	0.183
$\aleph(\mathscr{g}_{\Omega_2})$	0.199	0.000	0.174	0.210	0.166
$\aleph(\mathscr{g}_{\Omega_3})$	0.181	0.174	0.000	0.126	0.102
$\aleph(\mathscr{g}_{\Omega_4})$	0.215	0.210	0.126	0.000	0.156
$\aleph(\mathscr{g}_{\Omega_5})$	0.183	0.166	0.102	0.156	0.000
Trigonal Bipyramidal	$\aleph(\mathscr{g}_{\Omega_1})$	$\aleph(\mathscr{g}_{\Omega_2})$	$\aleph(\mathscr{g}_{\Omega_3})$	$\aleph(\mathscr{g}_{\Omega_4})$	$\aleph(\mathscr{g}_{\Omega_5})$
$\aleph(\mathscr{g}_{\Omega_1})$	0.000	0.208	0.189	0.225	0.191
$\aleph(\mathscr{g}_{\Omega_2})$	0.208	0.000	0.182	0.219	0.173
$\aleph(\mathscr{g}_{\Omega_3})$	0.189	0.182	0.000	0.131	0.106
$\aleph(\mathscr{g}_{\Omega_4})$	0.225	0.219	0.131	0.000	0.163
$\aleph(\mathscr{g}_{\Omega_5})$	0.191	0.173	0.106	0.163	0.000
Octahedral	$\aleph(\mathscr{g}_{\Omega_1})$	$\aleph(\mathscr{g}_{\Omega_2})$	$\aleph(\mathscr{g}_{\Omega_3})$	$\aleph(\mathscr{g}_{\Omega_4})$	$\aleph(\mathscr{g}_{\Omega_5})$
$\aleph(\mathscr{g}_{\Omega_1})$	0.000	0.242	0.220	0.262	0.222
$\aleph(\mathscr{g}_{\Omega_2})$	0.242	0.000	0.212	0.255	0.202
$\aleph(\mathscr{g}_{\Omega_3})$	0.220	0.212	0.000	0.153	0.124
$\aleph(\mathscr{g}_{\Omega_4})$	0.262	0.255	0.153	0.000	0.190
$\aleph(\mathscr{g}_{\Omega_5})$	0.222	0.202	0.124	0.190	0.000

Table A29
The reciprocal values for linear shape.

	$\Im(\mathscr{g}_{\Omega_1})$	$\Im(\mathscr{g}_{\Omega_2})$	$\Im(\mathscr{g}_{\Omega_3})$	$\Im(\mathscr{g}_{\Omega_4})$	$\Im(\mathscr{g}_{\Omega_5})$
$\Im(\mathscr{g}_{\Omega_1})$		8.271	9.078	7.638	9.009
$\Im(\mathscr{g}_{\Omega_2})$	8.271		9.451	7.835	9.915
$\Im(\mathscr{g}_{\Omega_3})$	9.078	9.451		13.091	16.187
$\Im(\mathscr{g}_{\Omega_4})$	7.638	7.835	13.091		10.513
$\Im(\mathscr{g}_{\Omega_5})$	9.009	9.915	16.187	10.513	

Table A30
The final decision matrix.

	LFCYAS	**DYSYSOT**	**GEOINFSYS**	**ECOSRV**	**ENVPER**
LFCYAS		0.082	0.090	0.076	0.089
DYSYSOT	0.082		0.094	0.078	0.098
GEOINFSYS	0.090	0.094		0.130	0.160
ECOSRV	0.076	0.078	0.130		0.104
ENVPER	0.089	0.098	0.160	0.104	

Table A31
The iterative results of velocity, updated personal and global best positions.

	LFCYAS $((\boldsymbol{V}_1)(1))$	**DYSYSOT** $((\boldsymbol{V}_2)(1))$	**GEOINFSYS** $((\boldsymbol{V}_3)(1))$	**ECOSRV** $((\boldsymbol{V}_4)(1))$	**ENVPER** $((\boldsymbol{V}_5)(1))$	**LFCYAS** $((\boldsymbol{S}_1)(1))$	**DYSYSOT** $((\boldsymbol{S}_2)(1))$	**GEOINFSYS** $((\boldsymbol{S}_3)(1))$	**ECOSRV** $((\boldsymbol{S}_4)(1))$	**ENVPER** $((\boldsymbol{S}_5)(1))$	$((\boldsymbol{S}_{gb})(1))$	$((\boldsymbol{S}_{gb})(\boldsymbol{t}))-((\boldsymbol{S}_{gb})(\boldsymbol{t}-1))$
LFCYAS		0.00	0.01	0.01	0.00	0.00	0.09	0.09	0.10	0.09	0.09	–
DYSYSOT	−0.01		0.00	−0.01	−0.01	0.10	0.00	0.10	0.10	0.10	0.10	–
GEOINFSYS	0.01	0.01		0.00	0.01	0.18	0.18	0.00	0.17	0.16	0.16	–
ECOSRV	0.00	−0.01	0.00		0.01	0.14	0.14	0.13	0.00	0.14	0.13	–
ENVPER	0.00	0.00	0.01	0.01		0.17	0.18	0.16	0.18	0.00	0.16	–
	LFCYAS $((\boldsymbol{V}_1)(2))$	**DYSYSOT** $((\boldsymbol{V}_2)(2))$	**GEOINFSYS** $((\boldsymbol{V}_3)(2))$	**ECOSRV** $((\boldsymbol{V}_4)(2))$	**ENVPER** $((\boldsymbol{V}_5)(2))$	**LFCYAS** $((\boldsymbol{S}_1)(2))$	**DYSYSOT** $((\boldsymbol{S}_2)(2))$	**GEOINFSYS** $((\boldsymbol{S}_3)(2))$	**ECOSRV** $((\boldsymbol{S}_4)(2))$	**ENVPER** $((\boldsymbol{S}_5)(2))$	$((\boldsymbol{S}_{gb})(2))$	$((\boldsymbol{S}_{gb})(\boldsymbol{t}))-((\boldsymbol{S}_{gb})(\boldsymbol{t}-1))$
LFCYAS		0.01	0.01	0.00	0.01	0.00	0.10	0.10	0.10	0.10	0.10	0.01
DYSYSOT	0.00		0.02	0.00	0.01	0.11	0.00	0.11	0.11	0.11	0.11	0.01
GEOINFSYS	0.02	0.03		0.02	0.04	0.21	0.21	0.00	0.20	0.19	0.19	0.03
ECOSRV	0.01	0.01	0.01		0.03	0.17	0.16	0.15	0.00	0.16	0.15	0.02
ENVPER	0.03	0.01	0.02	0.03		0.21	0.20	0.18	0.21	0.00	0.18	0.02
	LFCYAS $((\boldsymbol{V}_1)(3))$	**DYSYSOT** $((\boldsymbol{V}_2)(3))$	**GEOINFSYS** $((\boldsymbol{V}_3)(3))$	**ECOSRV** $((\boldsymbol{V}_4)(3))$	**ENVPER** $((\boldsymbol{V}_5)(3))$	**LFCYAS** $((\boldsymbol{S}_1)(3))$	**DYSYSOT** $((\boldsymbol{S}_2)(3))$	**GEOINFSYS** $((\boldsymbol{S}_3)(3))$	**ECOSRV** $((\boldsymbol{S}_4)(3))$	**ENVPER** $((\boldsymbol{S}_5)(3))$	$((\boldsymbol{S}_{gb})(3))$	$((\boldsymbol{S}_{gb})(\boldsymbol{t}))-((\boldsymbol{S}_{gb})(\boldsymbol{t}-1))$
LFCYAS		−0.02	−0.01	−0.03	0.00	0.00	0.10	0.10	0.10	0.10	0.10	0.00
DYSYSOT	−0.04		−0.01	−0.05	0.00	0.11	0.00	0.11	0.11	0.11	0.11	0.01
GEOINFSYS	−0.16	−0.14		−0.07	−0.01	0.21	0.21	0.00	0.20	0.19	0.21	0.03
ECOSRV	−0.12	−0.11	−0.01		−0.06	0.17	0.16	0.15	0.00	0.16	0.17	0.02
ENVPER	−0.16	−0.14	−0.01	−0.12		0.21	0.20	0.19	0.21	0.00	0.21	0.03
	LFCYAS $((\boldsymbol{V}_1)(4))$	**DYSYSOT** $((\boldsymbol{V}_2)(4))$	**GEOINFSYS** $((\boldsymbol{V}_3)(4))$	**ECOSRV** $((\boldsymbol{V}_4)(4))$	**ENVPER** $((\boldsymbol{V}_5)(4))$	**LFCYAS** $((\boldsymbol{S}_1)(4))$	**DYSYSOT** $((\boldsymbol{S}_2)(4))$	**GEOINFSYS** $((\boldsymbol{S}_3)(4))$	**ECOSRV** $((\boldsymbol{S}_4)(4))$	**ENVPER** $((\boldsymbol{S}_5)(4))$	$((\boldsymbol{S}_{gb})(4))$	$((\boldsymbol{S}_{gb})(\boldsymbol{t}))-((\boldsymbol{S}_{gb})(\boldsymbol{t}-1))$
LFCYAS		−0.03	−0.01	−0.05	−0.01	0.00	0.10	0.10	0.10	0.10	0.10	0.00
DYSYSOT	−0.06		−0.03	−0.07	−0.01	0.11	0.00	0.11	0.11	0.11	0.11	0.00
GEOINFSYS	−0.25	−0.24		−0.12	−0.04	0.21	0.21	0.00	0.20	0.20	0.21	0.00
ECOSRV	−0.19	−0.18	−0.02		−0.11	0.17	0.16	0.16	0.00	0.16	0.17	0.00
ENVPER	−0.26	−0.22	−0.03	−0.20		0.21	0.20	0.21	0.21	0.00	0.21	0.00

Data availability

No data was used for the research described in the article.

References

Acker, A., Chatterjee, E., Becker, L., Shutzer, M., & Capellini, N. (2023). Fossil fuels from extraction to emissions. In *The Routledge Handbook of Environmental History* (pp. 229–243). Routledge.

Adedoyin, F. F. (2024). Policies related to renewable energy–water–environment nexus. In *The Renewable Energy-Water-Environment Nexus* (pp. 423–439). Elsevier.

Ahakwa, I. (2024). Towards land degradation neutrality: Does green energy and green human capital matter? *Renewable and Sustainable Energy Reviews, 197*, Article 114396. https://doi.org/10.1016/j.rser.2024.114396

Akça, E. E. (2024). Do renewable energy sources improve air quality? Demand-and supply-side comparative evidence from industrialized and emerging industrial economies. *Environmental Science and Pollution Research, 31*(1), 293–311. https://doi.org/10.1007/s11356-023-30946-2

Arshed, N., Anwar, A., Abbas, M., & Mughal, W. (2024). Natural habitat vs human in competition for breathing space: Need for restructuring clean energy infrastructure. *Ecological Economics, 220*, Article 108177. https://doi.org/10.1016/j.ecolecon.2024.108177

Aydin, M., Guney, E., Yigit, B., Acikgoz, F., & Cakmak, B. Y. (2024). Regulatory pathways to green energy transition for sustainable environment: The fostering role of human rights, banking sector development, economic complexity, and economic freedom. *Journal of Environmental Management, 366*, Article 121739. https://doi.org/10.1016/j.jenvman.2024.121739

Basso, A., & Zolin, M. B. (2023). Analyzing the land and labour productivity of farms producing renewable energy: The Italian case study. *Journal of Productivity Analysis, 59*(2), 153–172. https://doi.org/10.1007/s11123-023-00659-2

Bukhary, S. S. (2024). The renewable energy–water nexus. In *The Renewable Energy-Water-Environment Nexus* (pp. 143-176). Elsevier. doi: 10.1016/B978-0-443-13439-5.00006-5.

Chen, X. H., Tee, K., Elnahass, M., & Ahmed, R. (2023). Assessing the environmental impacts of renewable energy sources: A case study on air pollution and carbon emissions in China. *Journal of Environmental Management, 345*, Article 118525. https://doi.org/10.1016/j.jenvman.2023.118525

Coppitters, D., & Contino, F. (2023). Optimizing upside variability and antifragility in renewable energy system design. *Scientific Reports, 13*(1), 9138. https://doi.org/10.1038/s41598-023-36379-8

Cui, Q., Zhang, X., Yang, L., Ali, T., Yang, Y., Wang, H., & Liu, R. (2025). Assessing decarbonization pathways by weighing carbon mitigation efficiency and risks in China's energy system. *Environmental Impact Assessment Review, 114*, Article 107935. https://doi.org/10.1016/j.eiar.2025.107935

Dagar, V., Rao, A., Dagher, L., & Doytch, N. (2024). Climate change dynamics for global energy security and equity: Evidence from policy stringency drivers. *Journal of Environmental Management, 370*, Article 122484. https://doi.org/10.1016/j.jenvman.2024.122484

Dias, A. M., Cook, C., Paglia, A. P., & Massara, R. L. (2025). Imperfect detection of terrestrial mammals in environmental impact assessment (EIA) baseline surveys may compromise decision-making and mitigation measures. *Environmental Impact Assessment Review, 112*, Article 107850. https://doi.org/10.1016/j.eiar.2025.107850

Edouard, S., Combes, D., Van Iseghem, M., Tin, M. N. W., & Escobar-Gutiérrez, A. J. (2023). Increasing land productivity with agriphotovoltaics: Application to an alfalfa field. *Applied Energy, 329*, Article 120207. https://doi.org/10.1016/j.apenergy.2022.120207

El-Sorogy, A. S., Al-Kahtany, K., Alharbi, T., Al Hawas, R., & Rikan, N. (2025). Geographic information system and multivariate analysis approach for mapping soil contamination and environmental risk assessment in arid regions. *Land, 14*(2), 221. https://doi.org/10.3390/land14020221

Gagliardi, G. G., Cosentini, C., Agati, G., Borello, D., & Venturini, P. (2025). Green hydrogen production via floating photovoltaic systems on irrigation reservoirs: An Italian case study. *Renewable Energy*. , Article 123040. https://doi.org/10.1016/j.renene.2025.123040

Gentili, P. L. (2024). The conformational contribution to molecular complexity and its implications for information processing in living beings and chemical artificial intelligence. *Biomimetics, 9*(2), 121. https://doi.org/10.3390/biomimetics9020121

Guo, Q., Shah, M. I., Kumar, S., AbdulKareem, H. K., & Inuwa, N. (2023). The roles of organic farming, renewable energy, and corruption on biodiversity crisis: A European perspective. *Environmental Science and Pollution Research, 30*(11), 31696–31710. https://doi.org/10.1007/s11356-022-24344-3

Gutsch, M., & Leker, J. (2024). Costs, carbon footprint, and environmental impacts of lithium-ion batteries–From cathode active material synthesis to cell manufacturing and recycling. *Applied Energy, 353*, Article 122132. https://doi.org/10.1016/j.apenergy.2023.122132

Hall, J., Zhong, J., Jowett, S., Mazzeo, A., Thomas, G. N., Bryson, J. R., & Bartington, S. E. (2024). Regional impact assessment of air quality improvement: The air quality lifecourse assessment tool (AQ-LAT) for the West Midlands combined authority (WMCA) area. *Environmental Pollution*. , Article 123871. https://doi.org/10.1016/j.envpol.2024.123871

Jafarinejad, S., Hernandez, R. R., Bigham, S., & Beckingham, B. S. (2023). The intertwined renewable energy–water–environment (REWE) nexus challenges and opportunities: A case study of California. *Sustainability, 15*(13), 10672. https://doi.org/10.3390/su151310672

Ji, Y., Zuo, Q., Zhao, C., Zhang, Z., & Wu, Q. (2025). Measurement and decomposition of green water resource utilization efficiency across multiple water use sectors in China: A perspective on water-saving potential. *Environmental Impact Assessment Review, 112*, Article 107806. https://doi.org/10.1016/j.eiar.2025.107806

Joof, F., Samour, A., Tursoy, T., & Ali, M. (2023). Climate change, insurance market, renewable energy, and biodiversity: Double-materiality concept from BRICS countries. *Environmental Science and Pollution Research, 30*(11), 28676–28689. https://doi.org/10.1007/s11356-022-24068-4

Kaczmarczyk, M., Mukti, M., Ghaffour, N., Soukane, S., Bundschuh, J., & Tomaszewska, B. (2024). Renewable energy-driven membrane distillation in the context of life cycle assessment. *Renewable and Sustainable Energy Reviews, 192*, Article 114249. https://doi.org/10.1016/j.rser.2023.114249

Kartal, M. T., Shahbaz, M., Taşkın, D., Depren, S. K., & Ayhan, F. (2024). How are energy transition and energy-related R&D investments effective in enabling decarbonization? Evidence from Nordic Countries by novel WLMC model. *Journal of Environmental Management, 365*, Article 121664. https://doi.org/10.1016/j.jenvman.2024.121664

Kwilinski, A. (2024). Mapping global research on green energy and green investment: A comprehensive bibliometric study. *Energies, 17*(5), 1119. https://doi.org/10.3390/en17051119

Kyttä, V., Ghani, H. U., Lindfors, K., Heikkinen, J., & Palosuo, T. (2025). Integrating policy targets into product environmental impact assessments: A case study with Finnish agricultural products. *Cleaner Environmental Systems, 16*, Article 100252. https://doi.org/10.1016/j.cesys.2024.100252

Li, Q., Chen, W., Shi, H., & Zhang, S. (2024). Assessing the environmental impact of agricultural production structure transformation—evidence from the non-grain production of cropland in China. *Environmental Impact Assessment Review, 106*, Article 107489. https://doi.org/10.1016/j.eiar.2024.107489

Naimoglu, M., Shahbaz, M., & Lorente, D. B. (2024). Road map from dirty growth to sustainable green growth in Turkiye: Management of trade and financial processes? *Journal of Environmental Management, 367*, Article 121984. https://doi.org/10.1016/j.jenvman.2024.121984

Noh, Y., Jafarinejad, S., & Beckingham, B. S. (2024). Technology development in the nexus of renewable energy, water, and the environment. In. In *The renewable energy-water-environment Nexus* (pp. 257–313). Elsevier. doi: 10.1016/B978-0-443-13439-5.00009-0.

Qin, Y., Zhou, M., Hao, Y., Huang, X., Tong, D., Huang, L., & Zhu, T. (2024). Amplified positive effects on air quality, health, and renewable energy under China's carbon neutral target. *Nature Geoscience, 1–8*. https://doi.org/10.1038/s41561-024-01425-1

Qing, L., Abbas, J., Najam, H., & Ma, X. (2024). Investment in renewable energy and green financing and their role in achieving carbon-neutrality and economic sustainability: Insights from Asian region. *Renewable Energy, 221*, Article 119830. https://doi.org/10.1016/j.renene.2023.119830

Rabbani, M., Hosseini, A., Karim, M. A., Fahimi, A., Karimi, K., & Vahidi, E. (2024). Environmental impact assessment of a novel third-generation biorefinery approach for astaxanthin and biofuel production. *Science of the Total Environment, 912*, Article 168733. https://doi.org/10.1016/j.scitotenv.2023.168733

Raihan, A. (2023). The dynamic nexus between economic growth, renewable energy use, urbanization, industrialization, tourism, agricultural productivity, forest area, and carbon dioxide emissions in the Philippines. *Energy Nexus, 9*, Article 100180. https://doi.org/10.1016/j.nexus.2023.100180

Sekhar, S. J., Samuel, M. S., Glivin, G., Le, T. G., & Mathimani, T. (2024). Production and utilization of green ammonia for decarbonizing the energy sector with a discrete focus on Sustainable Development Goals and environmental impact and technical hurdles. *Fuel, 360*, Article 130626. https://doi.org/10.1016/j.fuel.2023.130626

Shah, M. I., Usman, M., Obekpa, H. O., & Abbas, S. (2023). Nexus between environmental vulnerability and agricultural productivity in BRICS: What are the roles of renewable energy, environmental policy stringency, and technology? *Environmental Science and Pollution Research, 30*(6), 15756–15774. https://doi.org/10.1007/s11356-022-23179-2

Shanbhag, S. S., & Dixit, M. K. (2025). Integrating evolving energy economy scenarios in dynamic life cycle assessment of buildings. *Environmental Impact Assessment Review, 112*, Article 107772. https://doi.org/10.1016/j.eiar.2024.107772

Sharma, A., Mahajan, P., & Garg, R. (2024). End-of-life solar photovoltaic panel waste management in India: Forecasting and environmental impact assessment. *International Journal of Environmental Science and Technology, 21*(2), 1961–1980. https://doi.org/10.1007/s13762-023-04953-2

Sivalingam, S., Vishal, G., & Anush, B. (2024). Environmental and health effects of acid rain. In *Health and environmental effects of ambient air pollution* (pp. 91–107). Academic Press. https://doi.org/10.1016/B978-0-443-16088-2.00007-7.

Sleiman, S., & Ouf, M. (2025). A review on energy practices and indoor environmental quality (IEQ) of airports. *Building and Environment*. , Article 112965. https://doi.org/10.1016/j.buildenv.2025.112965

Soni, S., & RL, M. (2024). The dynamic nexus between agricultural productivity and renewable energy consumption in BRICS: The role of financial inclusion and foreign direct investment. *International Journal of Energy Sector Management, 18*(6), 1885–1904. https://doi.org/10.1108/IJESM-08-2023-0028

Soto, G. H. (2024). Long run renewable energy productivity, carbon capture patents and air quality in Taiwan. *Journal of Environmental Management, 351*, Article 119925. https://doi.org/10.1016/j.jenvman.2023.119925

Su, S., Ju, J., Yuan, J., Chang, Y., & Li, Q. (2025). Interactive and dynamic insights into environmental impacts of a neighborhood: A tight coupling of multi-agent system and dynamic life cycle assessment. *Environmental Impact Assessment Review, 110*, Article 107708. https://doi.org/10.1016/j.eiar.2024.107708

Truong, T. K., To, T. T. L., Yoon, C. M., Kim, J. H., Park, Y. D., Youn, H., & Rho, J. (2025). Highly stretchable, robust, and floatable broadband light absorber of carbon nanotubes–ecoflex composite on 3D mogul-pattern for photothermal conversion. *Energy Technology*. , Article 2402169. https://doi.org/10.1002/ente.202402169

Wang, C., Ling, J., Liu, Y., Liu, B., & Deng, N. (2025). Can air quality ecological compensation improve environmental welfare performance? Based on the "Win–Win–Win" perspective of economy–ecology–welfare. *Journal of Cleaner Production, 489*, Article 144604. https://doi.org/10.1016/j.jclepro.2024.144604

Wang, C., Liu, Y., Zhang, X., Li, X., Paramygin, V., Sheng, P., & Xu, S. (2024). Scalable and rapid building damage detection after hurricane Ian using causal Bayesian networks and InSAR imagery. *International Journal of Disaster Risk Reduction, 104371*. https://doi.org/10.1016/j.ijdrr.2024.104371

Wang, J., & Zhang, Z. (2023). Impact of renewable energy and agriculture on mineral resources rents: Do economic and environmental aspects matter. *Resources Policy, 87*, Article 104281. https://doi.org/10.1016/j.resourpol.2023.104281

Xie, Y., Xu, M., Pu, J., Pan, Y., Liu, X., Zhang, Y., & Xu, S. (2023). Large-scale renewable energy brings regionally disproportional air quality and health co-benefits in China. *Iscience, 26*(8).

Yan, H., Qamruzzaman, M., & Kor, S. (2023). Nexus between green investment, fiscal policy, environmental tax, energy price, natural resources, and clean energy—a step towards sustainable development by fostering clean energy inclusion. *Sustainability, 15*(18), 13591. https://doi.org/10.3390/su151813591

Yi, D., Guo, J., Pueppke, S. G., Han, Y., Ding, G., Ou, M., & Koomen, E. (2025). What dominates the variation of ecosystem services across different urban expansion patterns?—Evidence from the Yangtze River Delta region, China. *Environmental Impact Assessment Review, 110*, Article 107674. https://doi.org/10.1016/j.eiar.2024.107674

Zhang, M., Ma, X., Mai, S., & Qing, L. (2025). Drivers of supply chain, environmental innovation, and digital population: The role of inflation and renewable energy on GHG emission in Indonesia. *Energy Economics*. , Article 108462. https://doi.org/10.1016/j.eneco.2025.108462

Zhou, Q., Lian, Y., Wu, J., Zhu, M., Wang, H., & Cao, J. (2024). An optimized Q-Learning algorithm for mobile robot local path planning. *Knowledge-Based Systems, 286*, Article 111400. https://doi.org/10.1016/j.knosys.2024.111400